

CS 3/5780: Introduction to Machine Learning

Lecture 1



John Thickstun



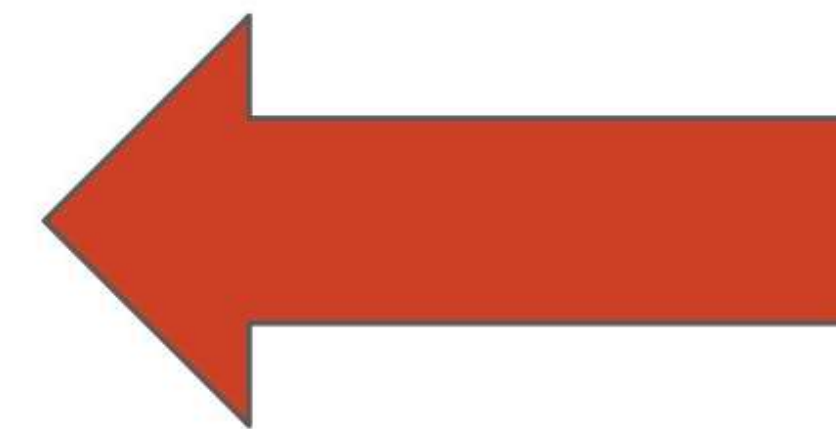
Karthik Sridharan

Do you like **video games?**
machine learning?



Help us evaluate a **game to teach machine learning!!**

<https://ela63-dgxspark.cs.cornell.edu/games/dot-partition/latest/?version=study>



Scan to play

optional

anonymized

~30 min

won't affect your grade

Course Links

- **Canvas:** <https://canvas.cornell.edu>
 - Most of the information about the course
 - Lectures notes, slides, references
 - For assignments, and connects to vocareum etc
- **Vocareum:** <http://vocareum.com>
 - Programming assignments are here
- **Ed:** <https://edstem.org/us/courses/69608/>
 - Discussion and questions

Where to get help?

- **Post all questions** on ED (fastest way to get answers)
 - For Emergency or privacy email: intro-ml-profs@cornell.edu
 - DO NOT Email instructors directly!
- **Instructor Office Hours:**
 - Karthik Sridharan: Wed 10:30am-11:30am, CIS 477
 - John Thickstun: Tue 3:00pm-4:00pm
- **TA Office Hours:** Details will follow soon...



Elaine, Wandering art museum



Lulu, Tennis



Guilia, Hiking and Rock climbing



Rose, Climbing (find me



Kevin, cooking new recipes ping pong



Nebil, Learning new languages



Danny, reading classic novels



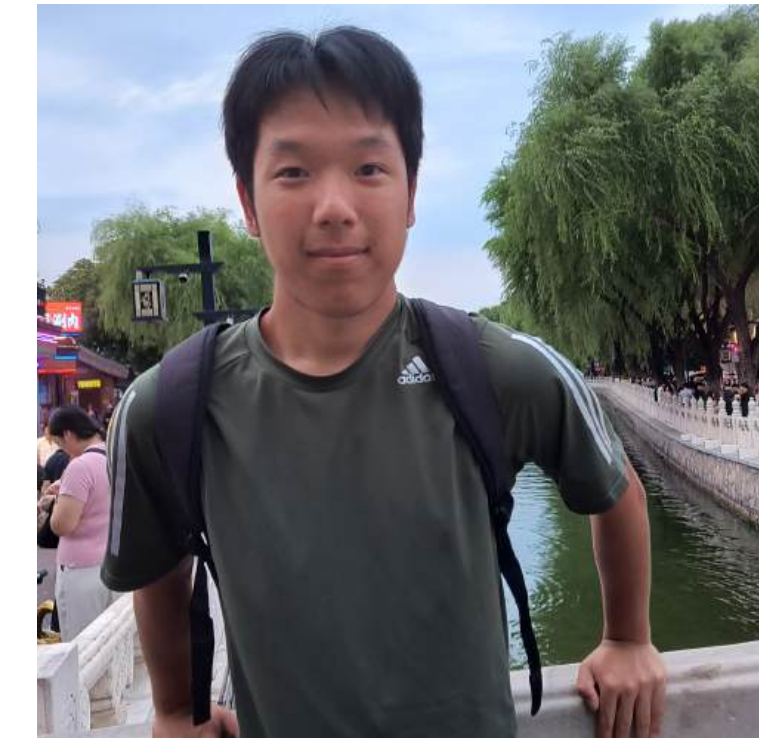
Leardon, reading



Alexander, video games



Amy, reading and baking



Jarmin, Hockey, chess, piano



Samruddhi, food spots, hot chocolate



Farhan, watching movies



Ashee, music, learning new instruments



Thurushi, painting, volleyball



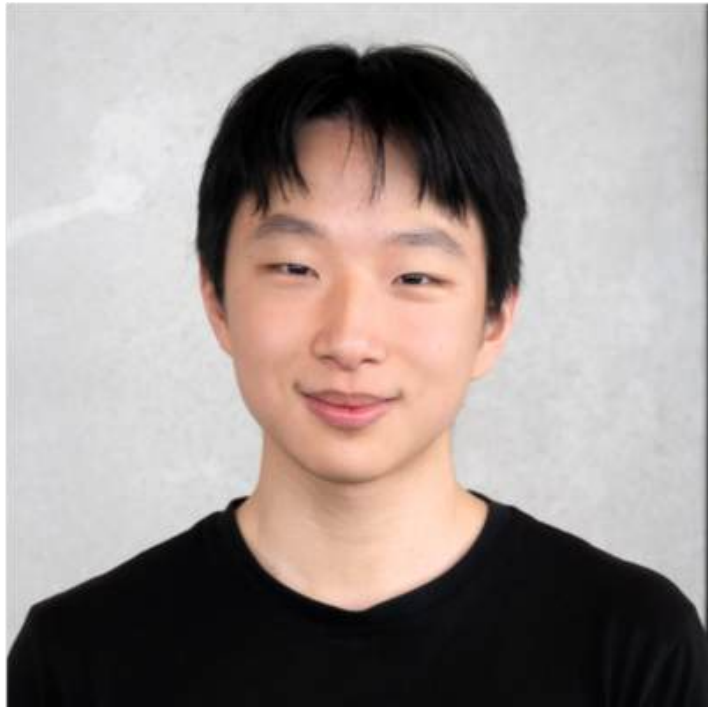
Justin, chess



Rohan, chess



Saesha, pickleball



Caleb, piano



Ilaria, whittling, rowing,
playing guitar



Jupiter, buying jewellery,
playing with cats



Boaz, piano and biking



Austin, volleyball,
snowboarding



John, running,
cooking



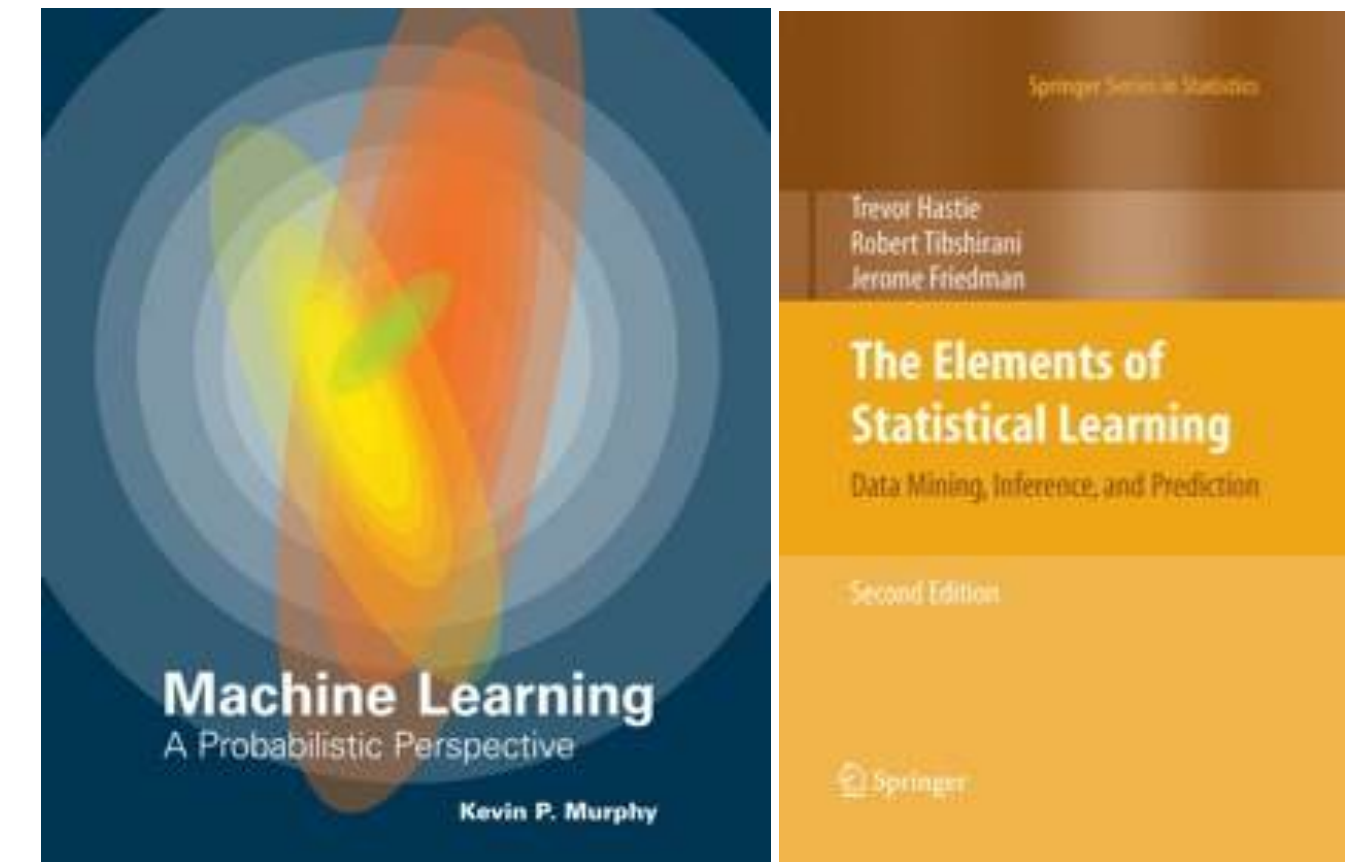
Karthik, rowing, playing the
flute

.....

References:

Main Course Texts

- Machine Learning a Probabilistic Perspective
 - K.P. Murphy
- The Elements of Statistical Learning
 - Hastie, Tibshirani, Friedman



Additional Texts

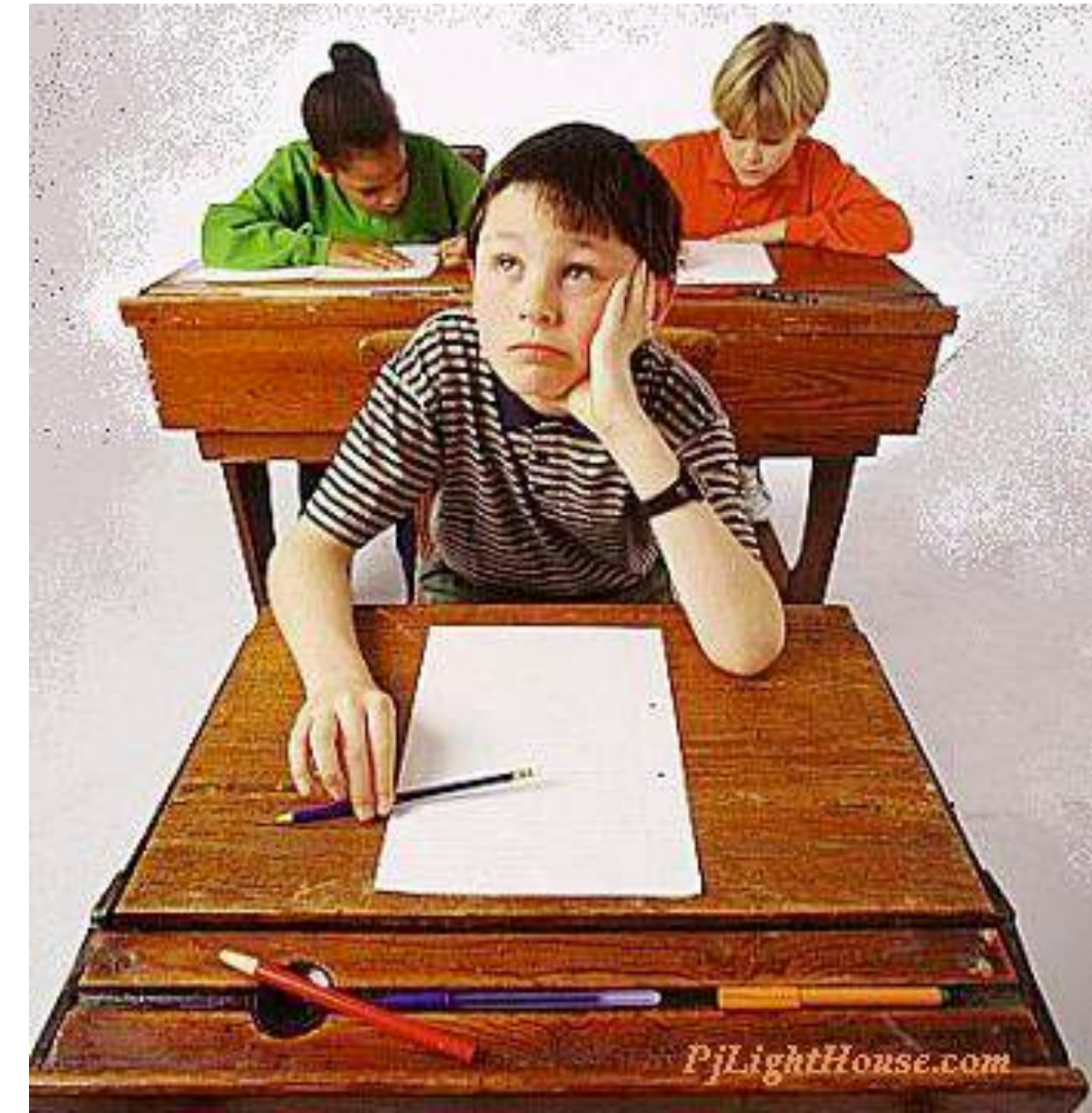
- An Introduction to Statistical Learning
 - James, Witten, Hastie, and Tibshirani
- Patterns, Predictions, and Actions
 - Hardt and Recht
- Fairness and Machine Learning
 - Barocas, Hardt, and Narayanan



Also: Background and programming resources on the website.

Placement Exam

- Due October 2nd 3pm (absolute deadline).
- **Canvas** page under Quiz.
- Get started early!
- This is how you get a Vocareum invite!
- Take it even if you are on the waitlist!
- If you are not on waitlist yet give us your names and netid end of this lecture so we can add you to canvas!

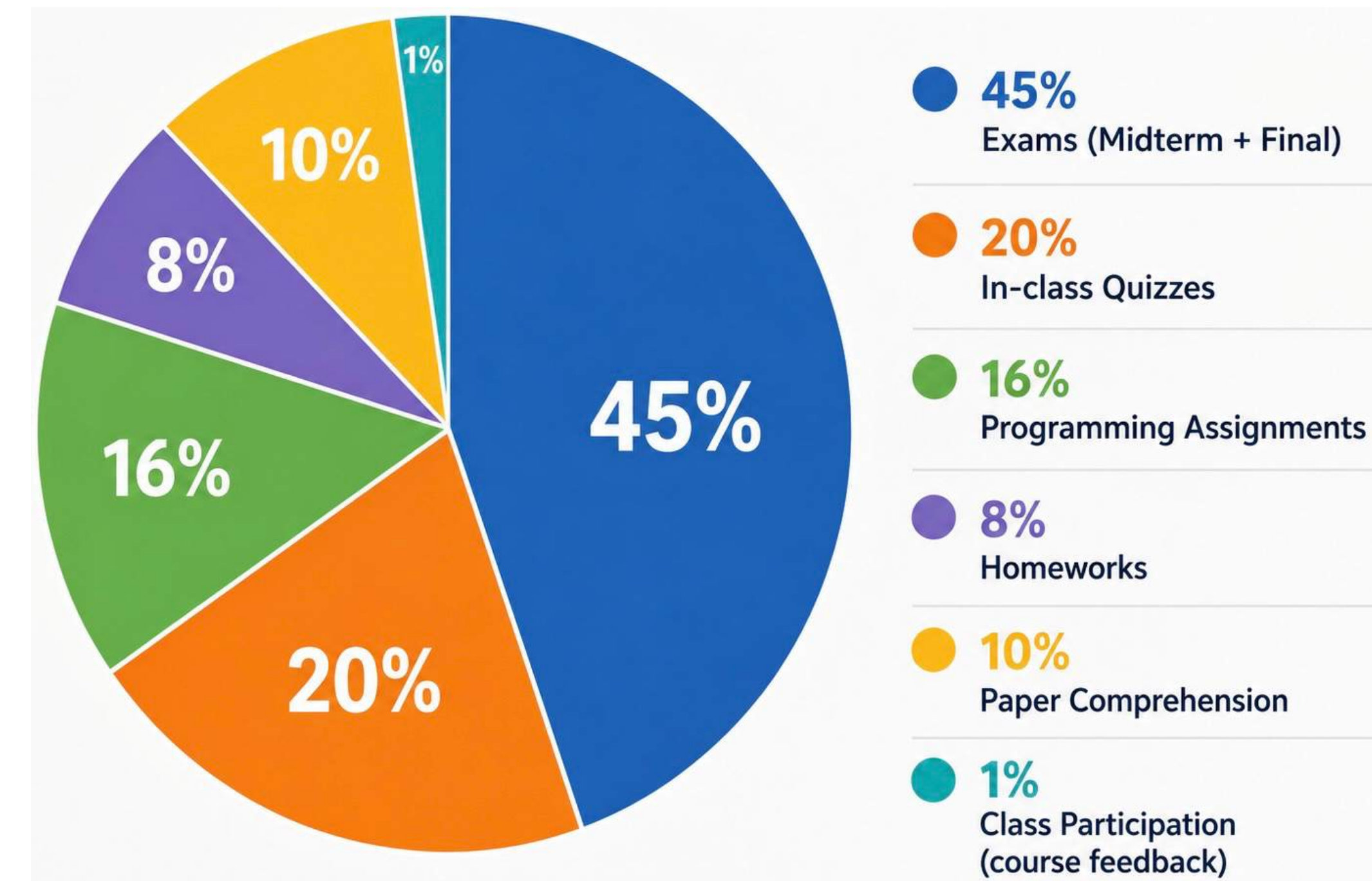


Course Components

- **Exams:** (1) Prelims and (2) Finals, Closed book
- **In class Quizzes:** Starting next week, every class, we drop the bottom 5!
- **Homework:** 7 in total
 - For reasonable effort completion only **no grades!**
 - Up to 4 members in each team
 - Prepares you for exam
- **Programming Projects: 8 (+2)**
 - Up to 2 members in each team
 - 2 days extension per team per project
 - Autograder (unlimited resubmissions)
 - Extra credit if you win contests

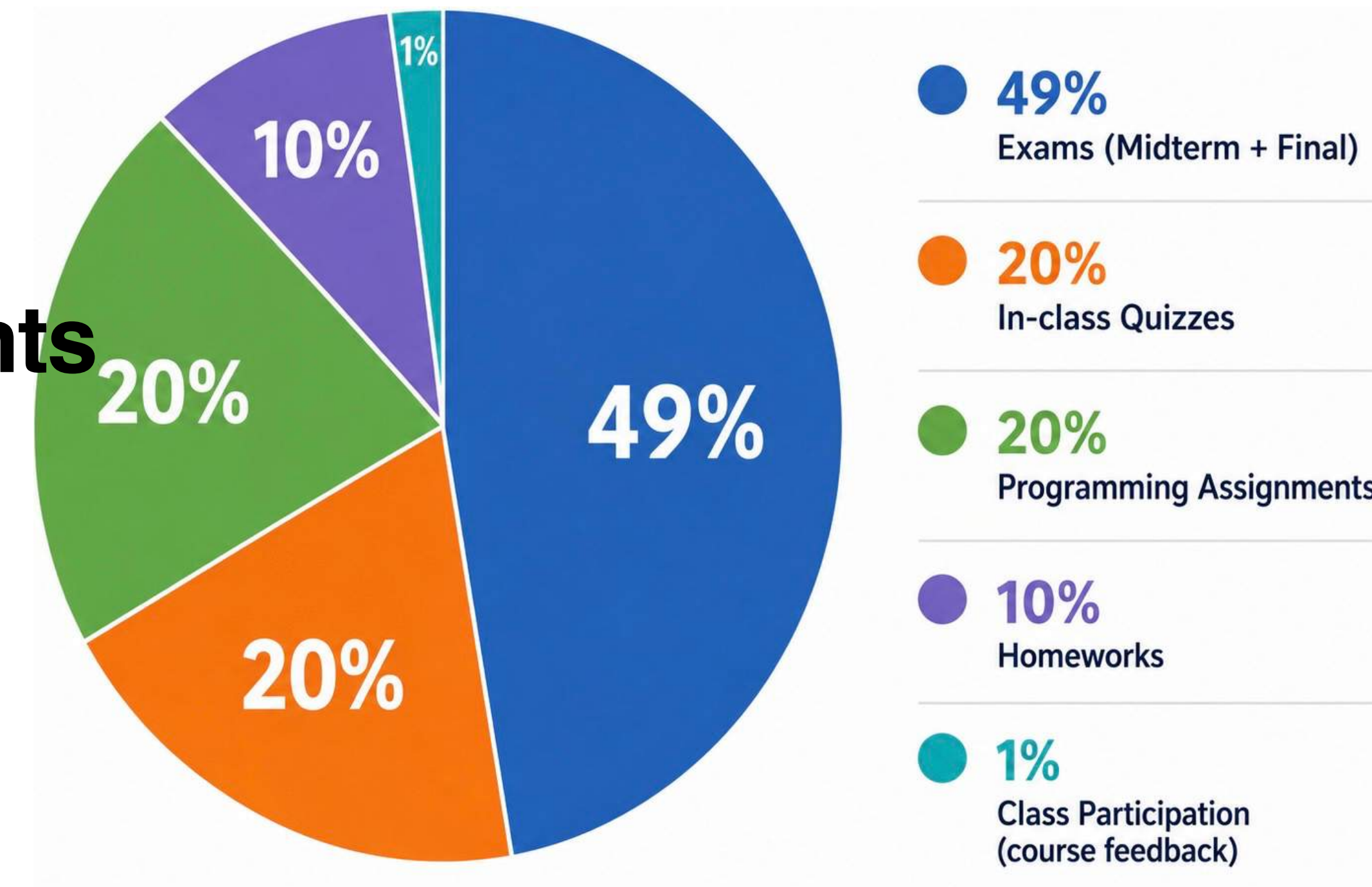
Grade Breakdown 5780

- **45% Exams**
- **20% In class Quizzes**
- **16% Programming Assignments**
- **8% Homeworks**
- **10% Paper Comprehension**



Grade Breakdown 3780

- **49% (or 45%) Exams**
- **20% In class Quizzes**
- **20% (or 16%) Programming Assignments**
- **10% (or 8%) Homeworks**

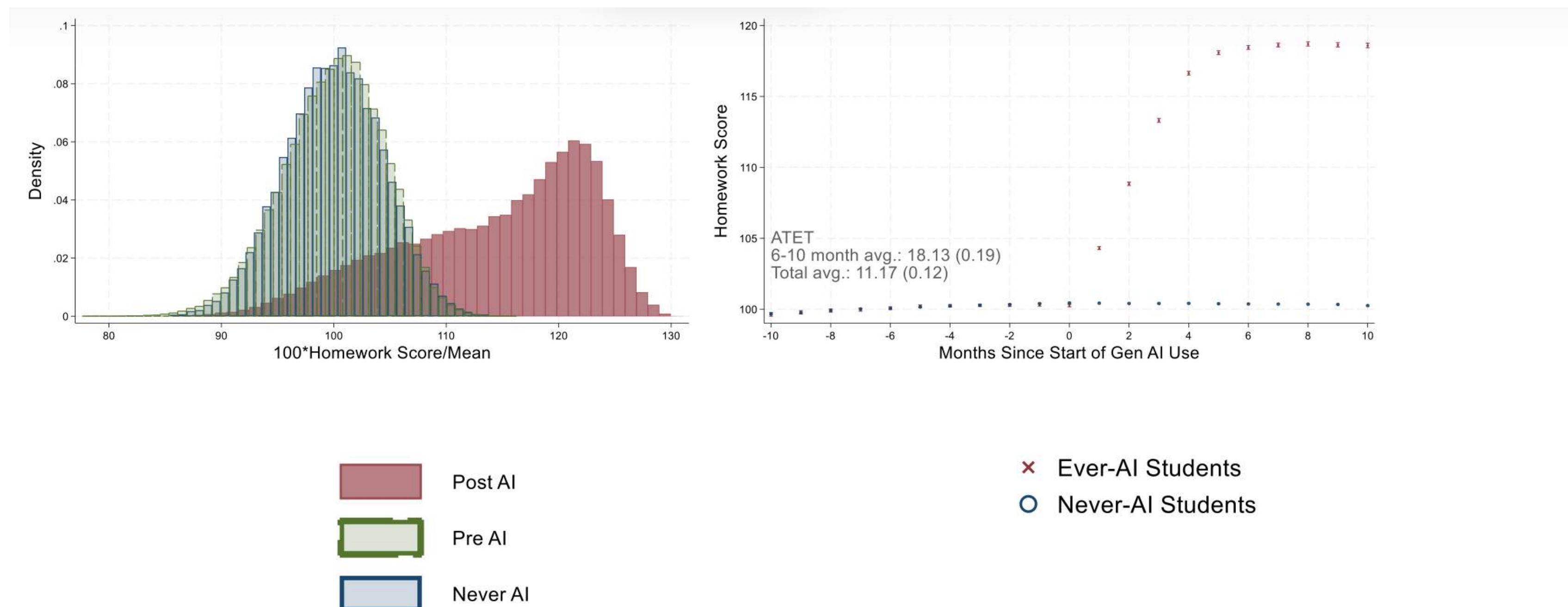


...or you can choose the
5780 grading scheme.

Learning in the Age of AI

- AI can be an incredible learning tool!
- Use it wisely:

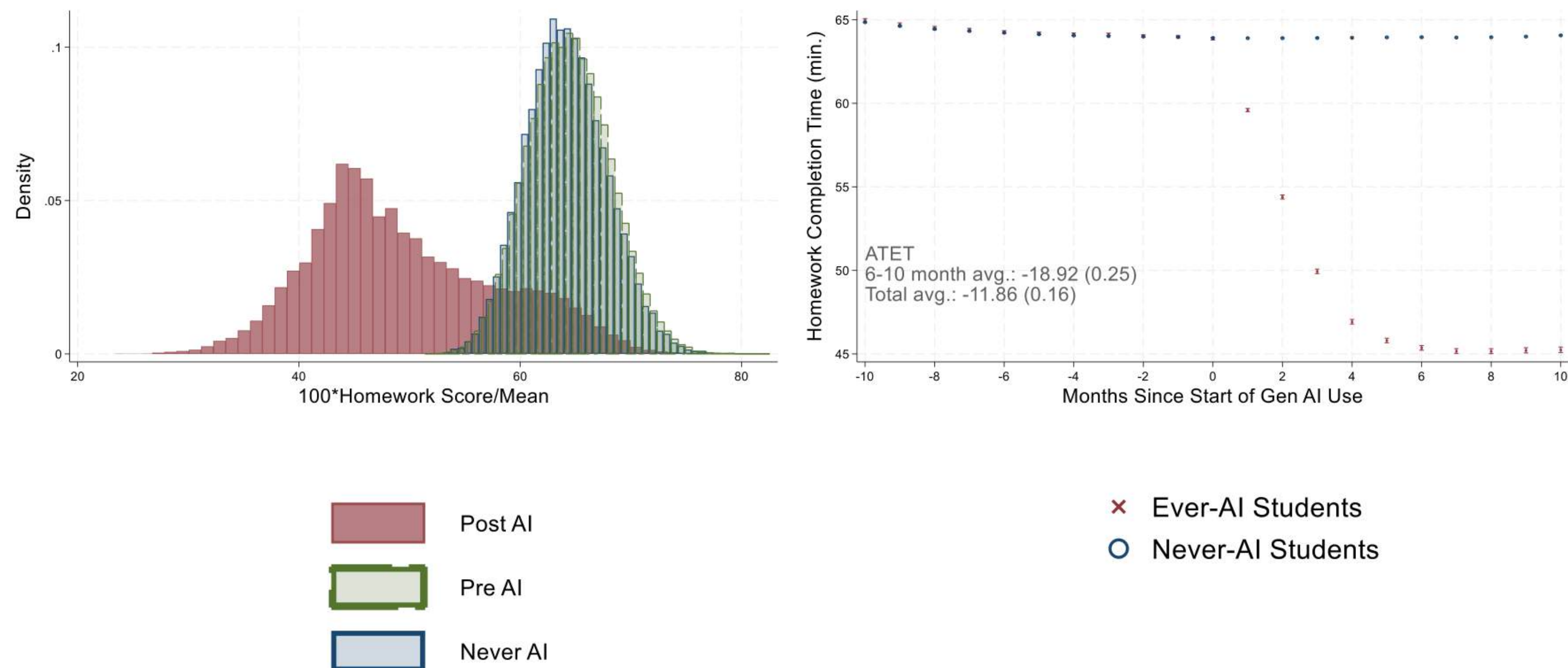
Homework Scores



Learning in the Age of AI

- AI can be an incredible learning tool!
- Use it wisely:

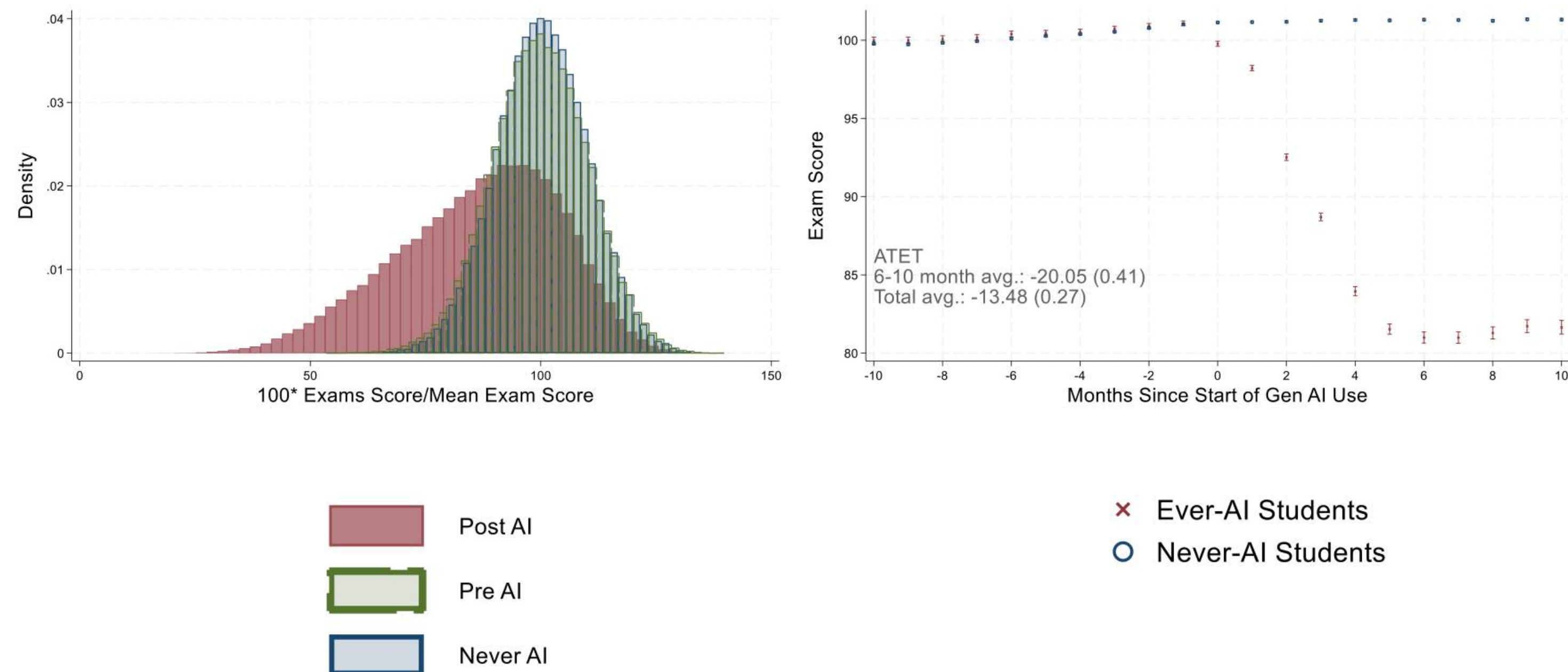
Homework Completion time



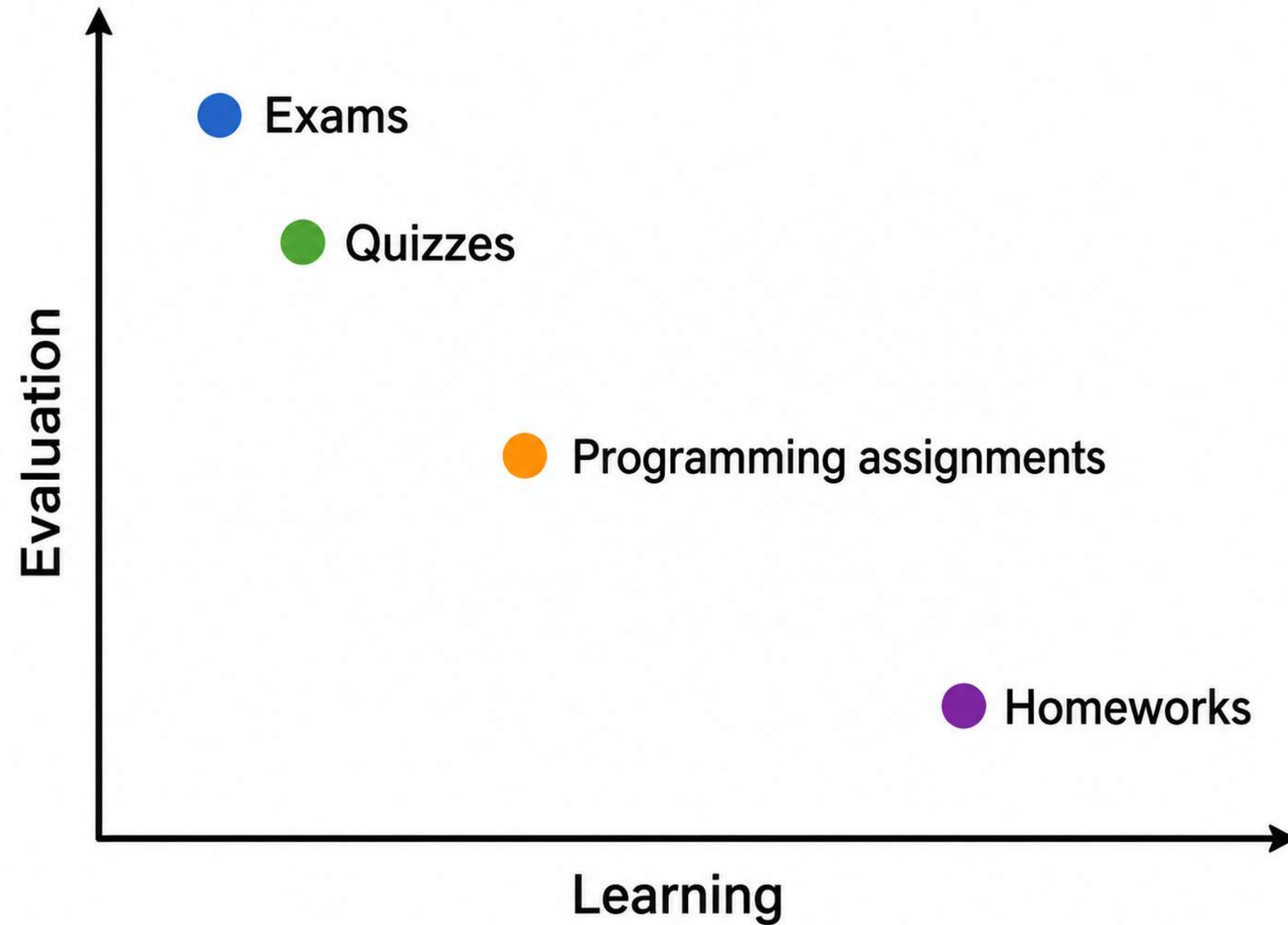
Learning in the Age of AI

- AI can be an incredible learning tool!
- Use it wisely:

Exam Scores



Learning Vs Evaluation



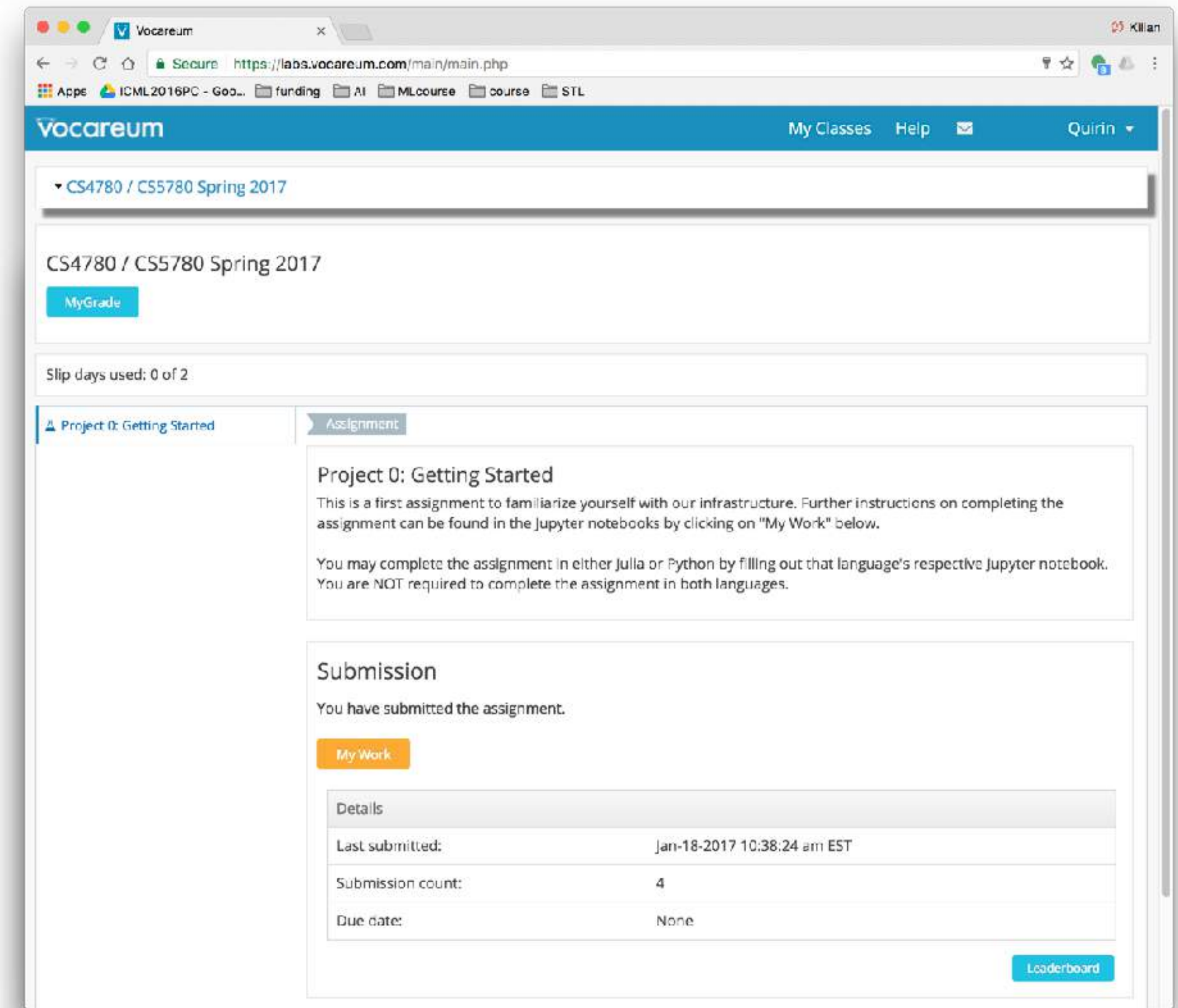
Study Groups (2–5 people)

- You **must** join a study group by the end of next week.
- Find people on **Ed**
- This course will cover non-trivial material, learning in a group makes it easier and more fun!



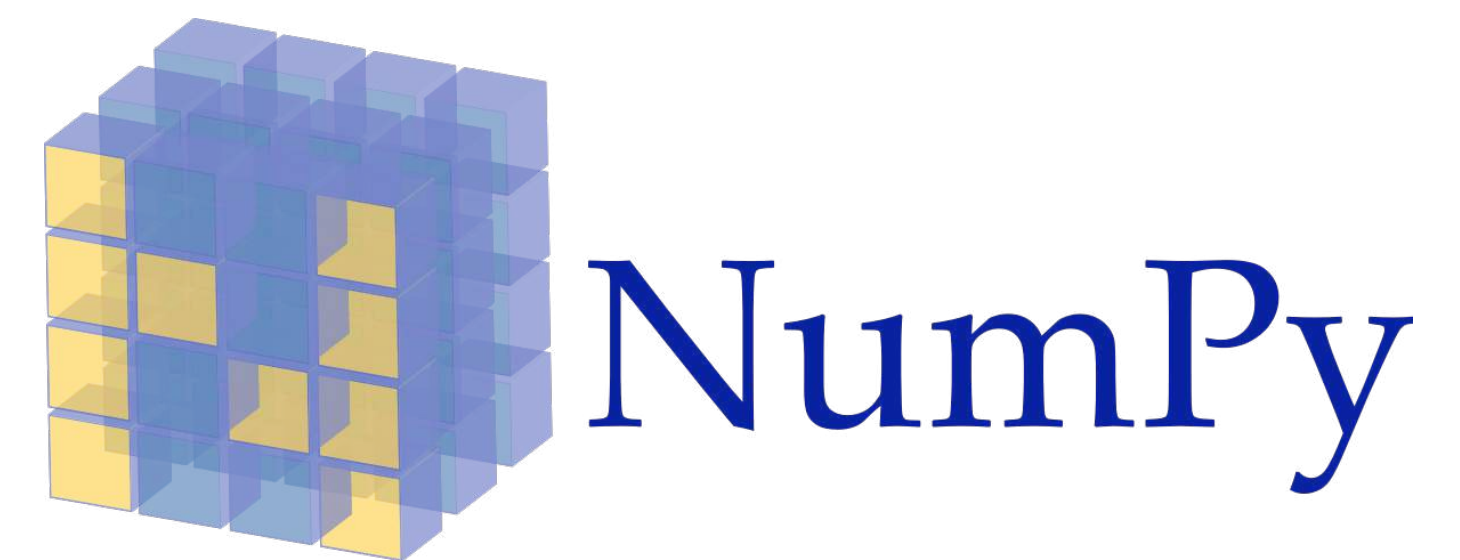
<http://vocareum.com>

- Pass placement exam → get account
- There will be 8(+2) projects
 - You have (roughly) 2 weeks for each project
 - Unlimited submits until deadline
- Costs \$30 :-()
- Group size at most 2



Placement Exam II (Project -1)

- NUMPY proficiency test
- Will turn into your own cheat-sheet
- Please take it seriously — this is for your own good
- We will go through the solutions in dedicated sections



Autograder

- Pick a (secret) name for your team for the leaderboard
 - (top right corner – click on your login)
- **Only text between #<GRADED> and #</GRADED> will be graded**
- Be sure to **form teams before you get started!**
 - Otherwise it creates headaches for the Vocareum system

Extra Credit

- For 3780 you can do paper reading + quizzes
- Kaggle competition that begins in the second half of the course and ends on last day of lectures is for extra credit only!

Course Topics

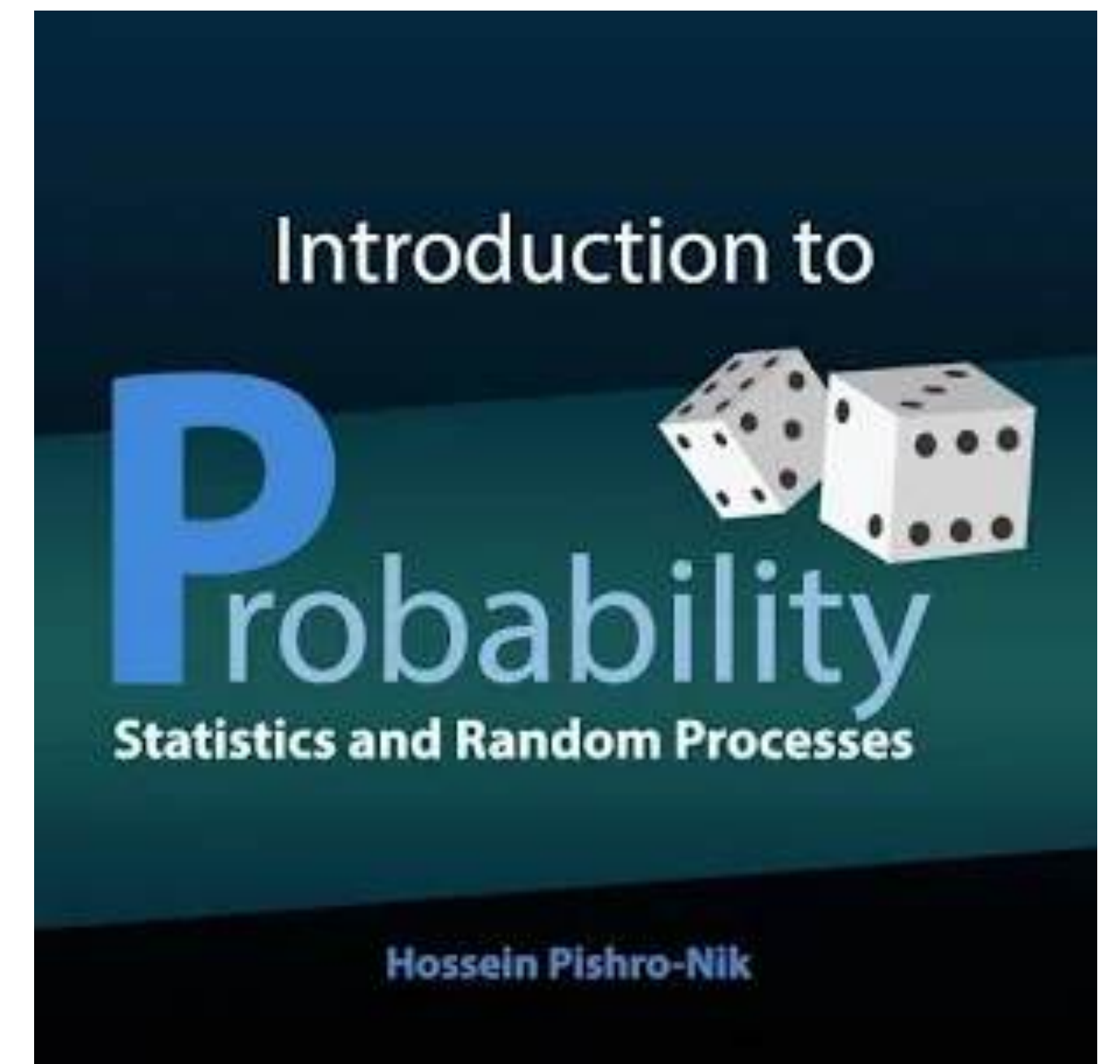
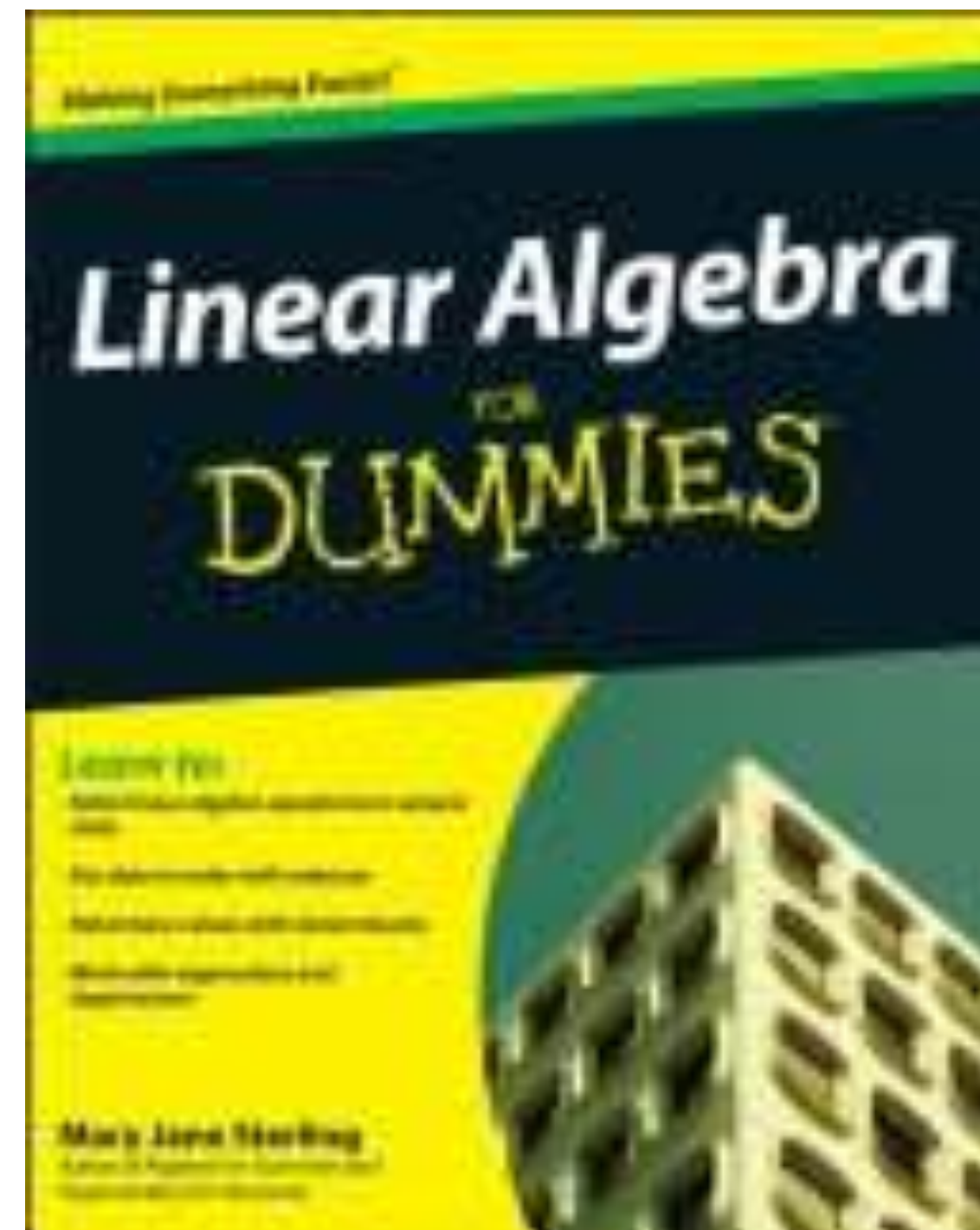
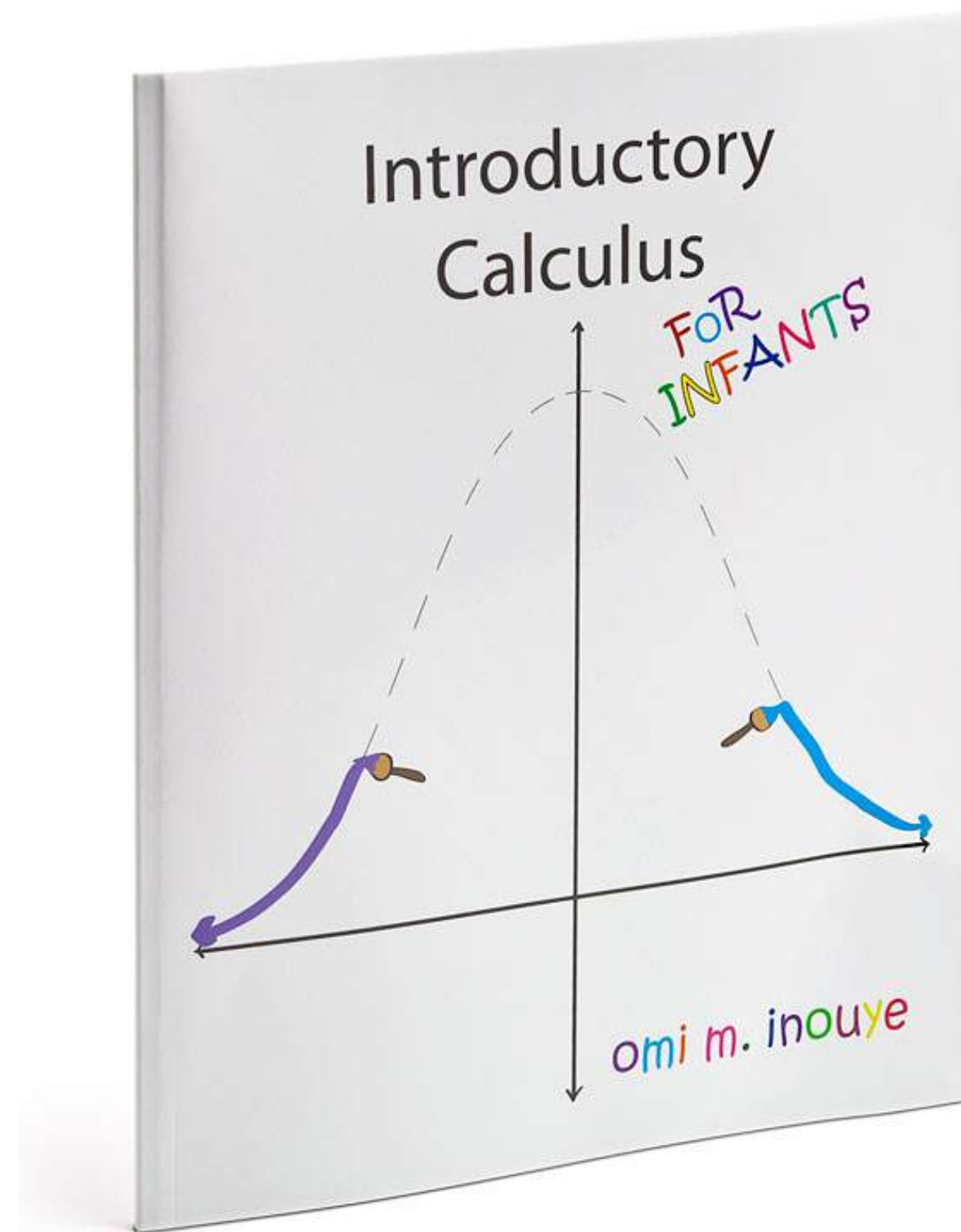
We will cover

- Parametric learning
- Non-parametric learning
- Empirical risk minimization
- Unsupervised learning
- Bias/Variance Trade-off
- Boosting
- Support Vector Machines
- Basic Deep Learning

We will not cover

- Graphical Models -> [CS3700]
- Reinforcement Learning -> [CS4789]
- Learning Theory -> [CS4783]
- More Deep Learning -> [CS4782]
- Genetic Programming

Prerequisites



Prerequisites

- Three pillars of ML:
 - **Statistics / Probability**
 - **Linear Algebra**
 - **Multivariate Calculus**
- Should be familiar with all, confident in at least $1/3$, ideally $2/3$.
- TAs might be able to give recitations on some topics if needed.
 - But don't rely on it.

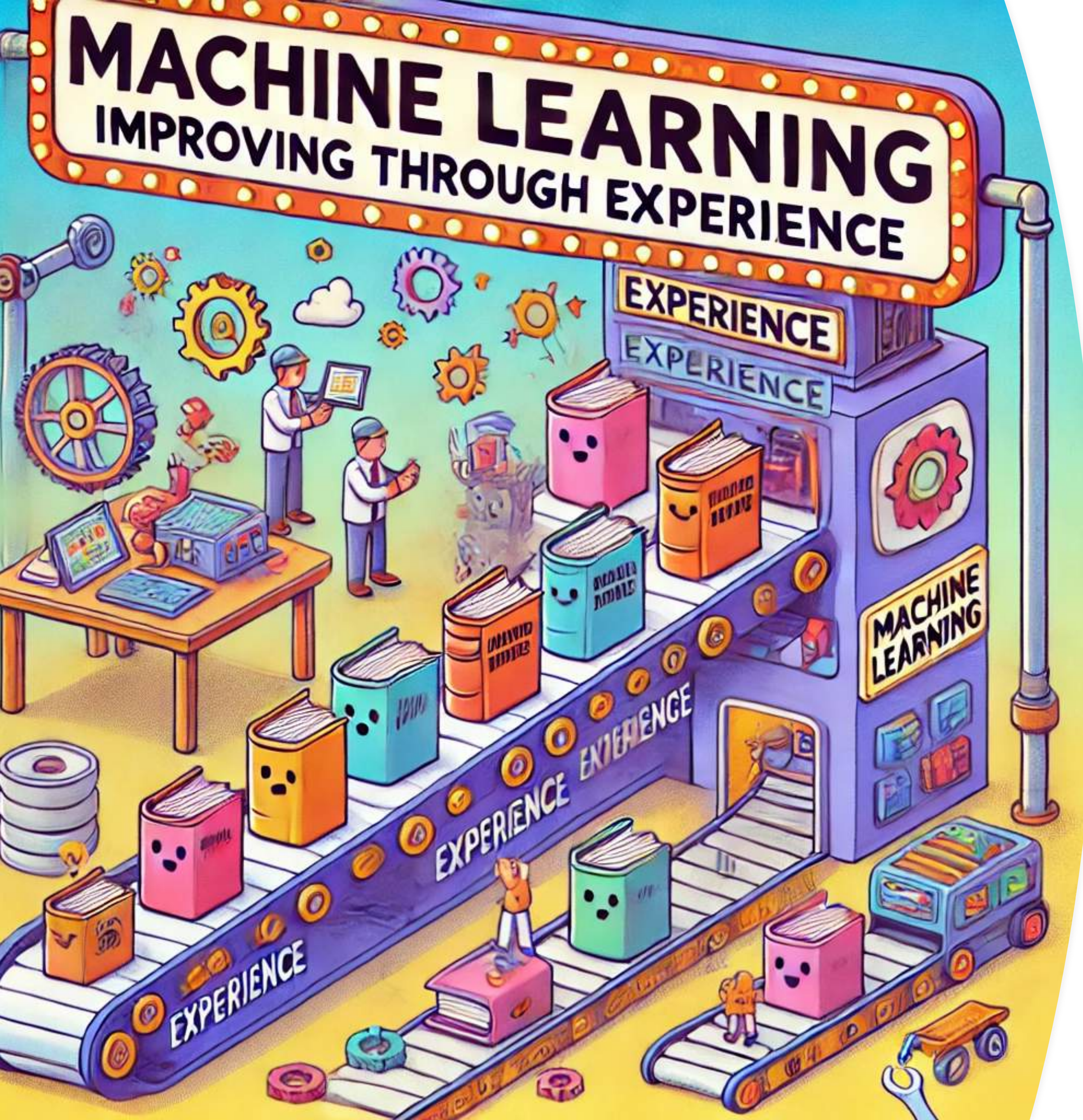


About this course

- **Take this course if ...**
 - you are interested in Machine Learning
 - you are comfortable with a decent amount of mathematics
 - you are proficient at programming
- **Don't take this course if ...**
 - matrices scare you
 - you don't remember how to take derivatives
 - you want an easy A
- You cannot take this course if you fail the placement exam.
 - In that case, take appropriate prerequisites and come back next year.

Working with AI

- This is a course about building AI. But you are all already users of AI!
- First try without AI. Struggling with hard ideas is how learning happens.
- Ask AI lots of questions. It is infinitely patient.
- The goal of homework is to help you build **personal understanding**.

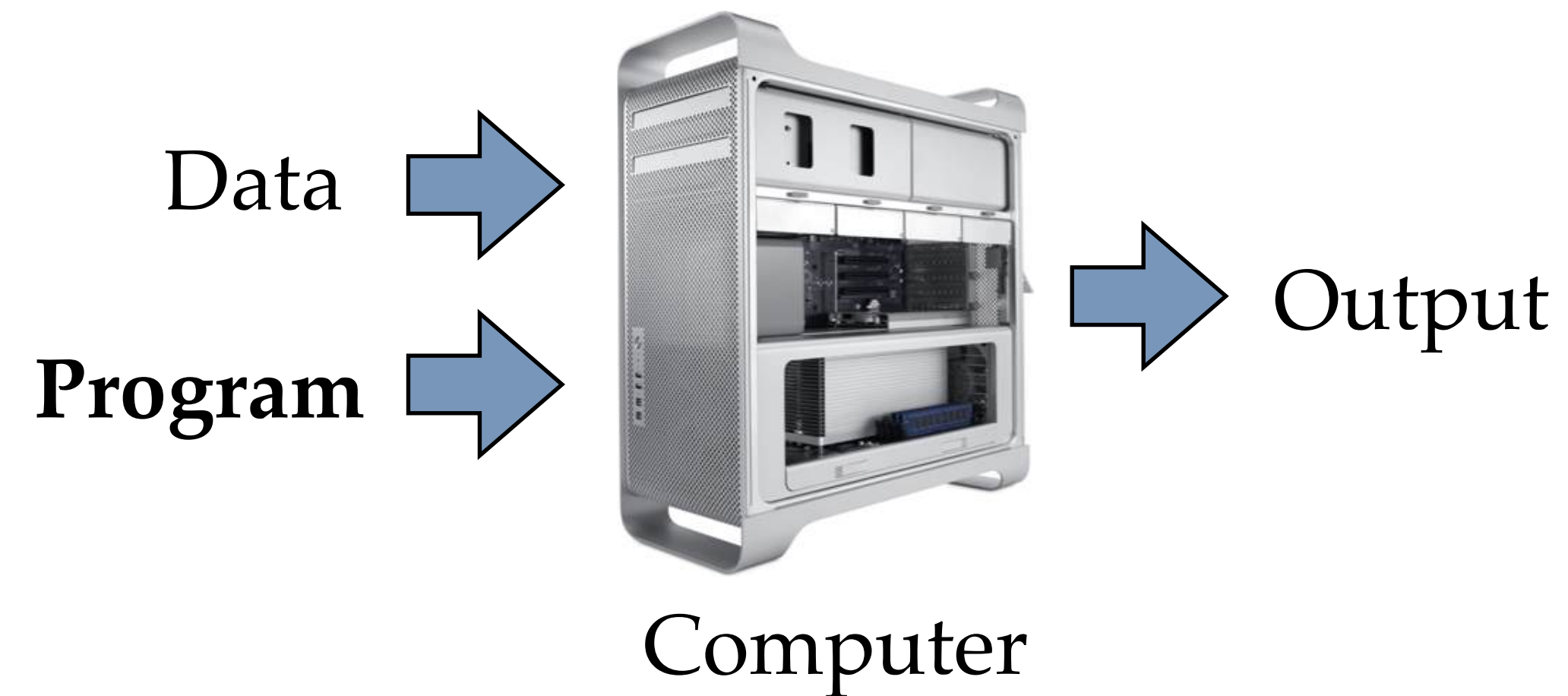


What is machine learning?

- Roughly: programs that **improve** with **experience**.
 - A subfield of **artificial intelligence**, which refers to programs that demonstrate “intelligence” in some sense

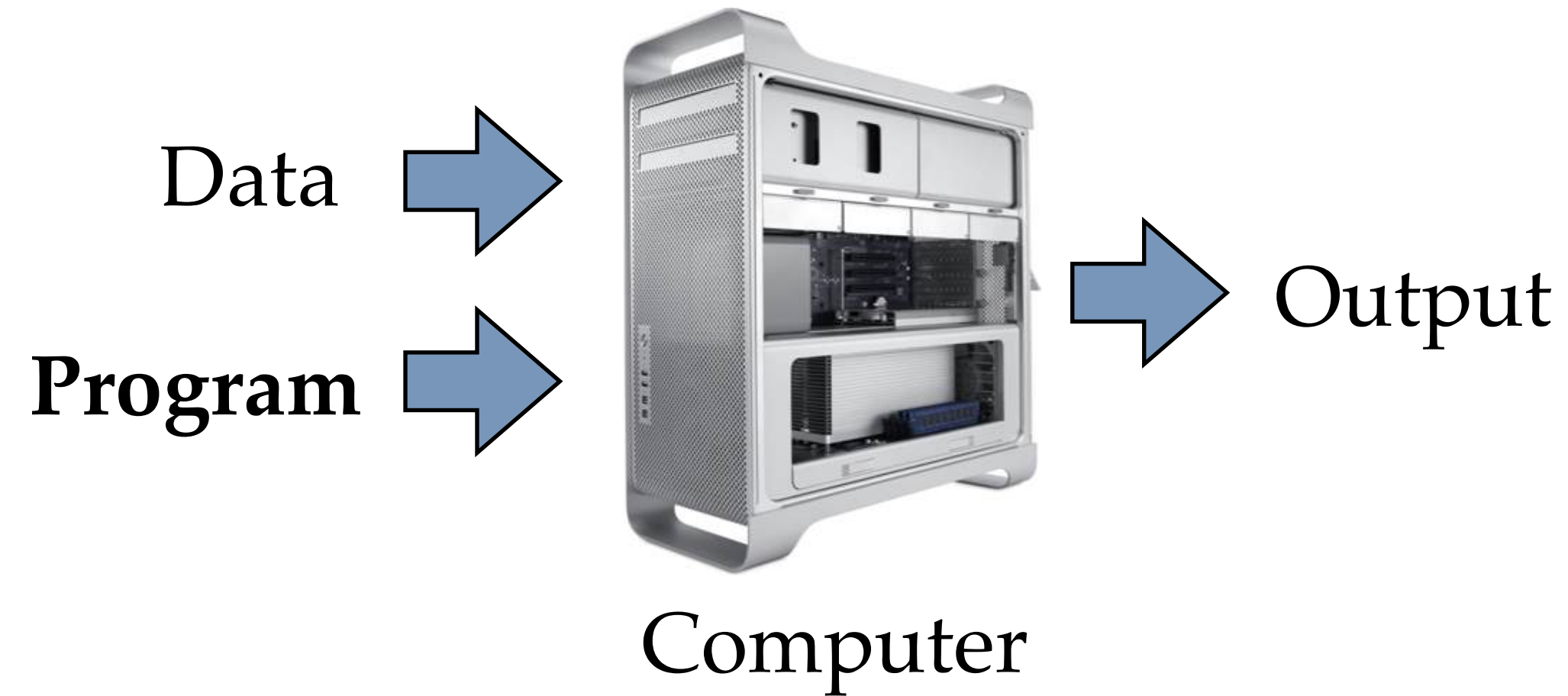
Traditional Computer Science

Traditional CS:

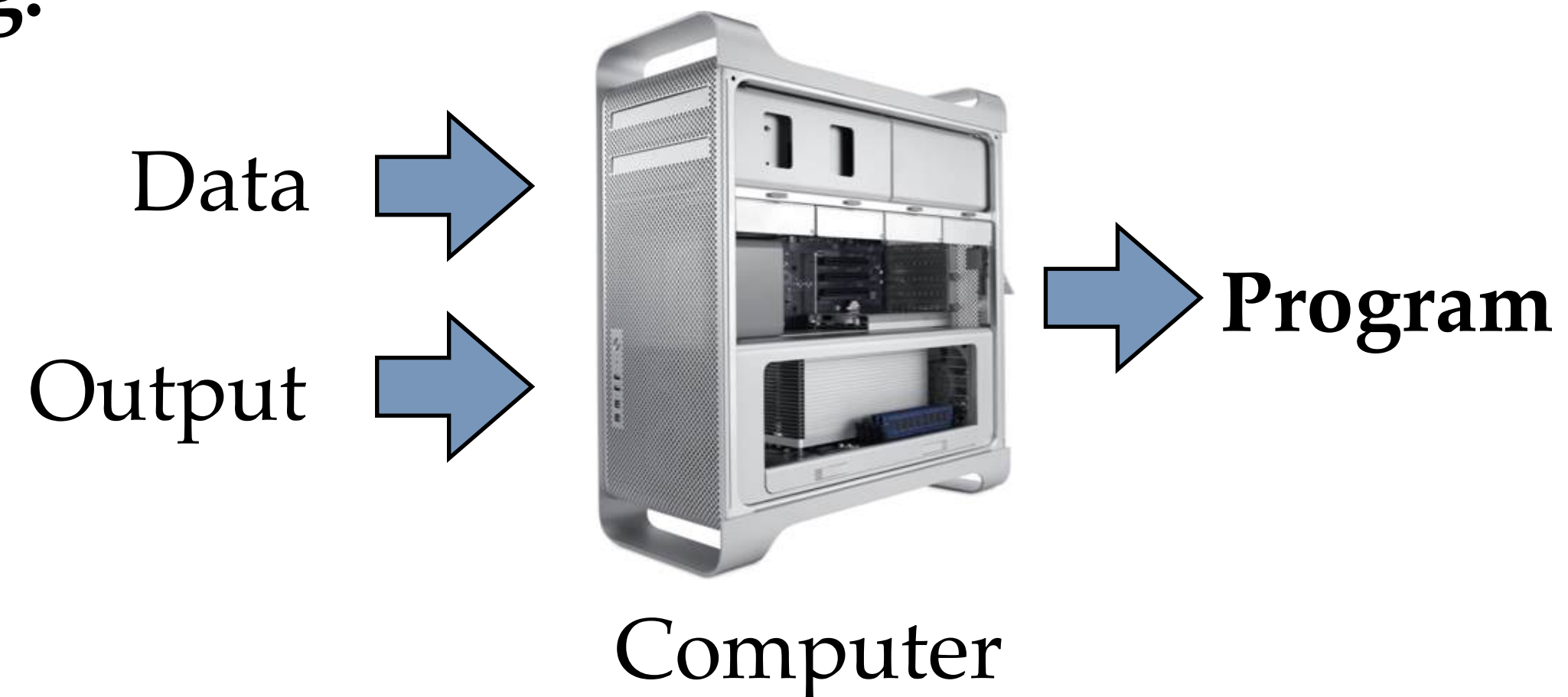


Machine Learning

Traditional CS:

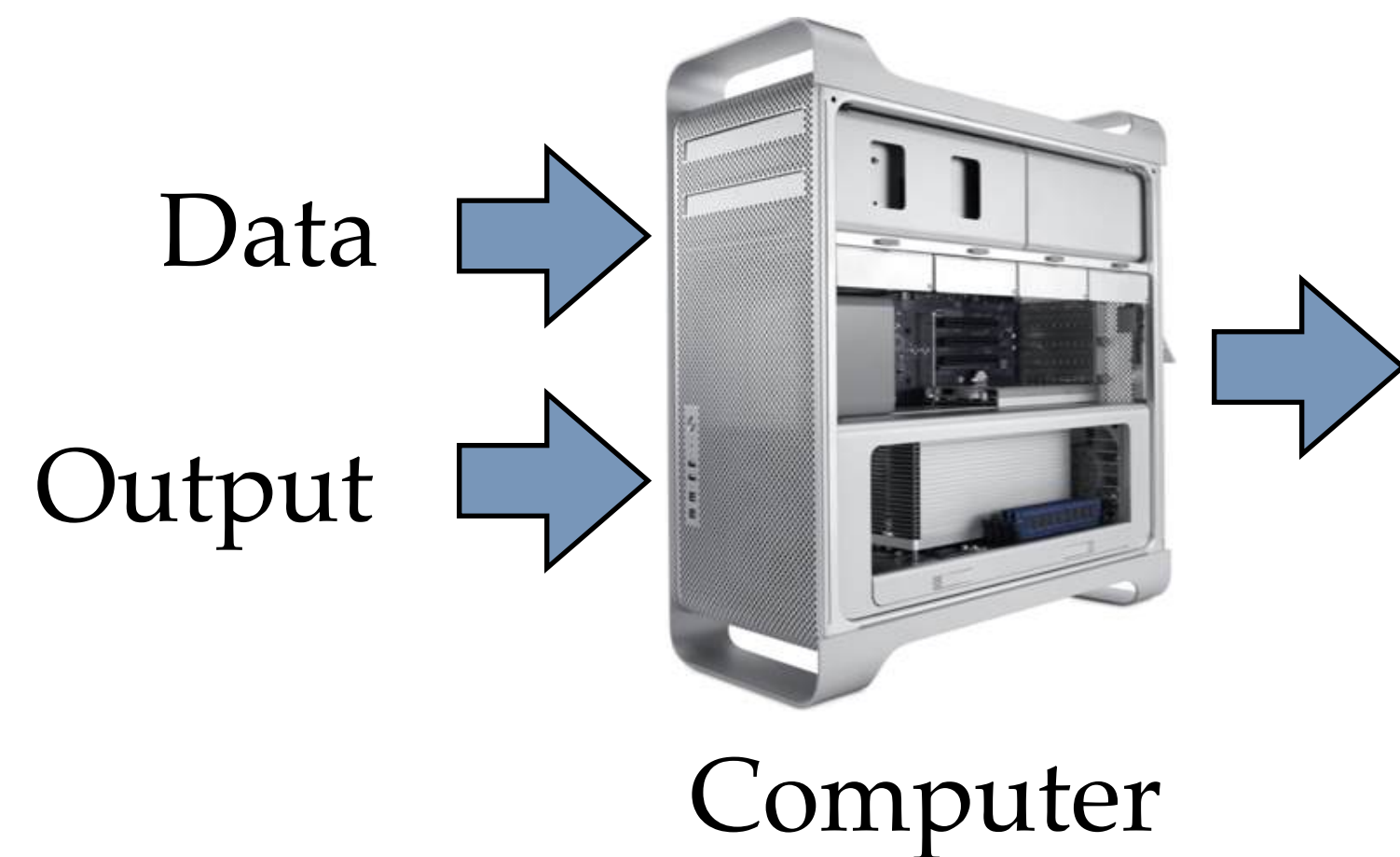


Machine Learning:

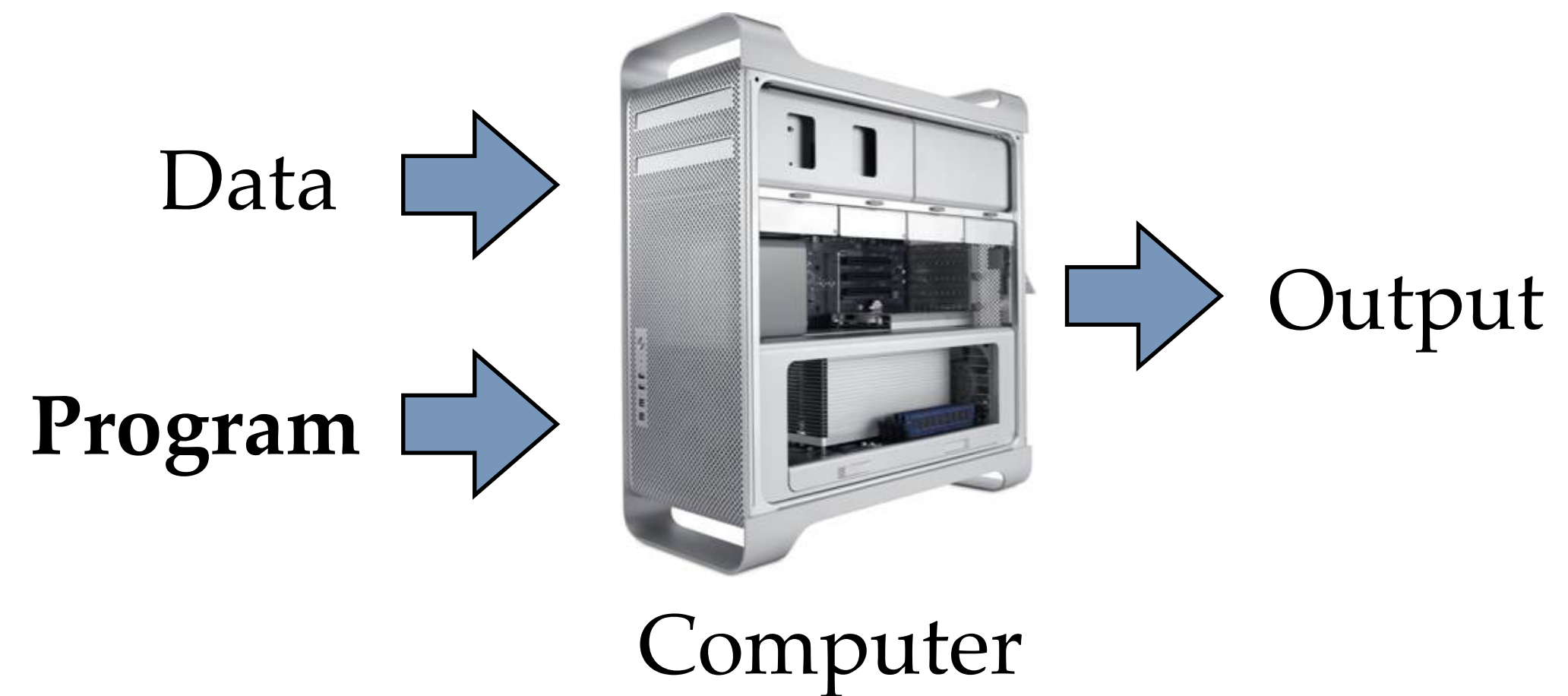


Machine Learning

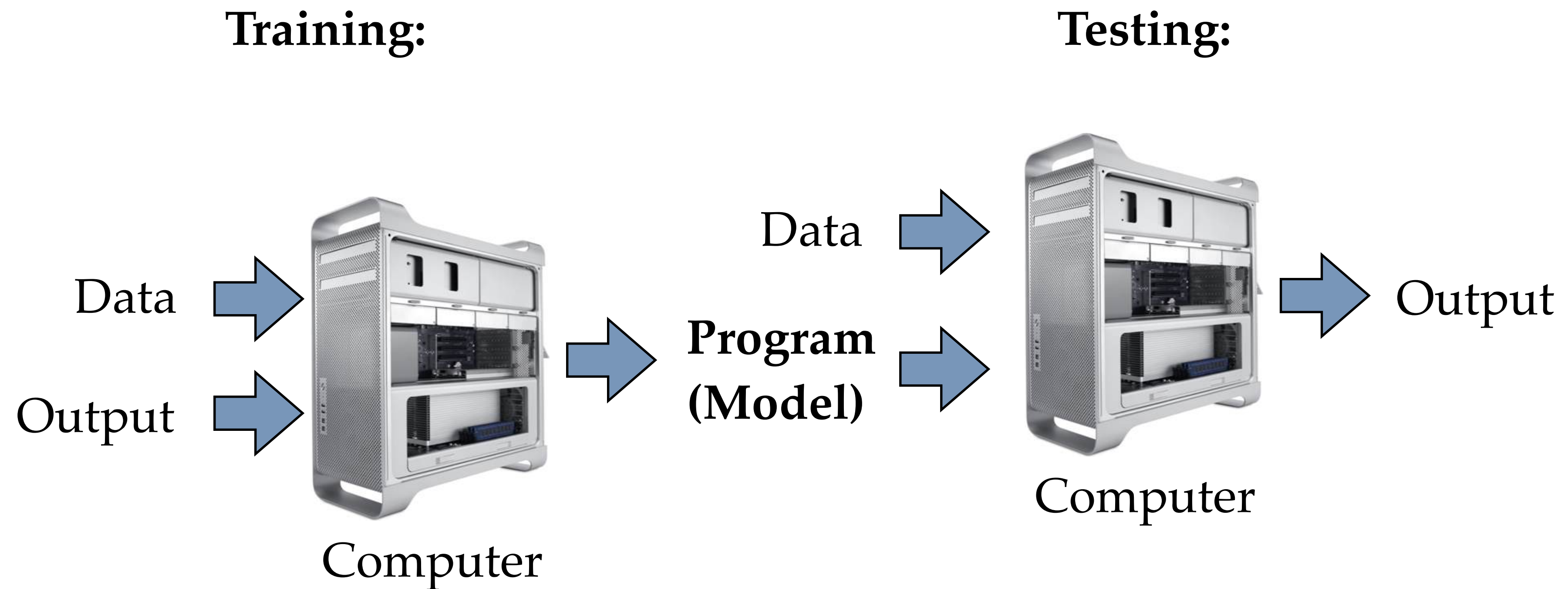
Machine Learning:

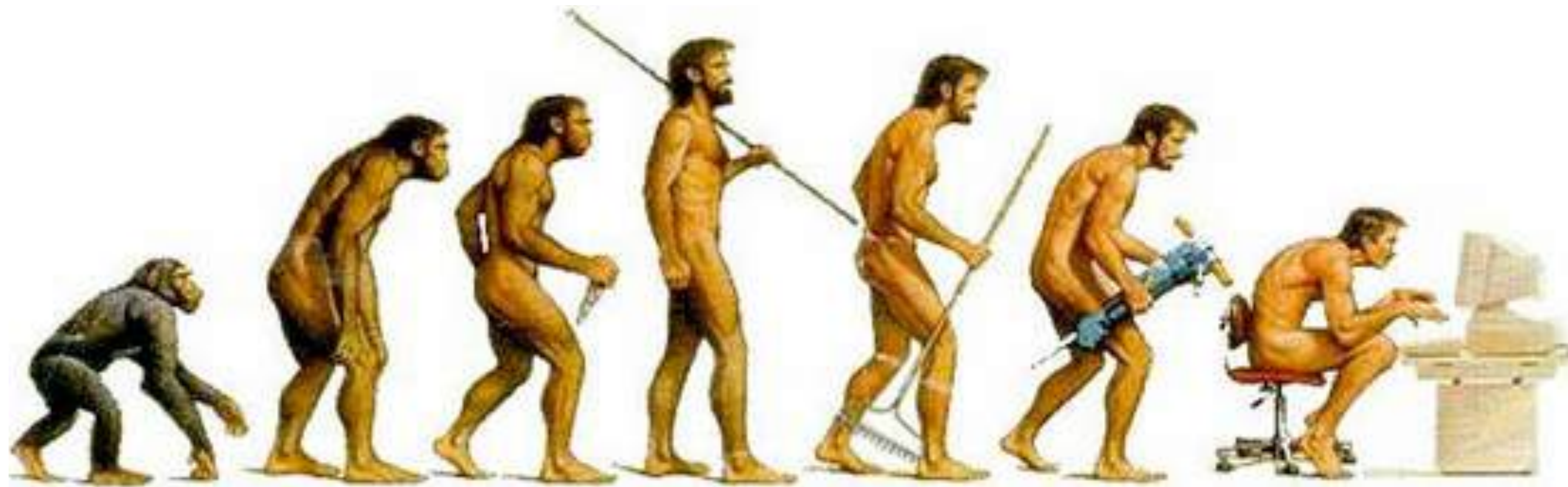


Traditional CS:



Machine Learning





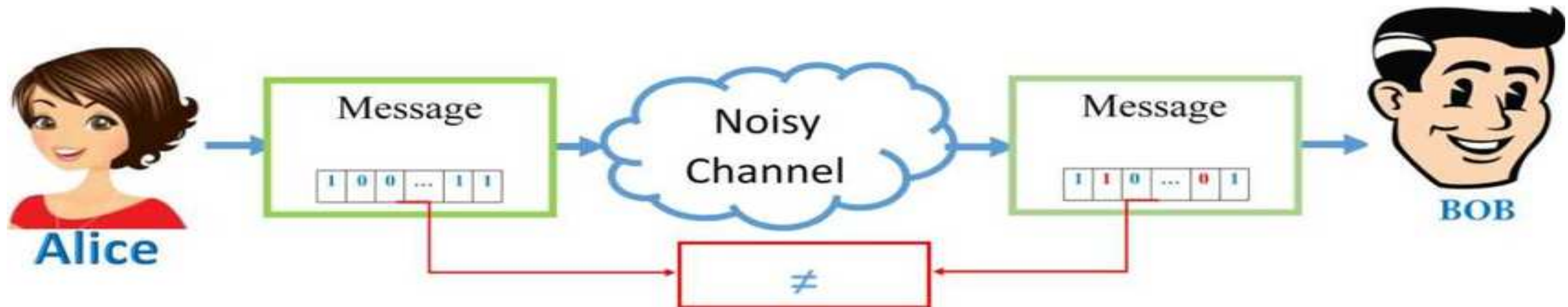
A (very brief) History of ML

A Mathematical Theory of Communication, 1948



Claude Shannon

Information can be *quantified* and *processed* using a digital computer.





Alan Turing

The Turing Test, 1950

A machine is intelligent if its answers are indistinguishable from a human's





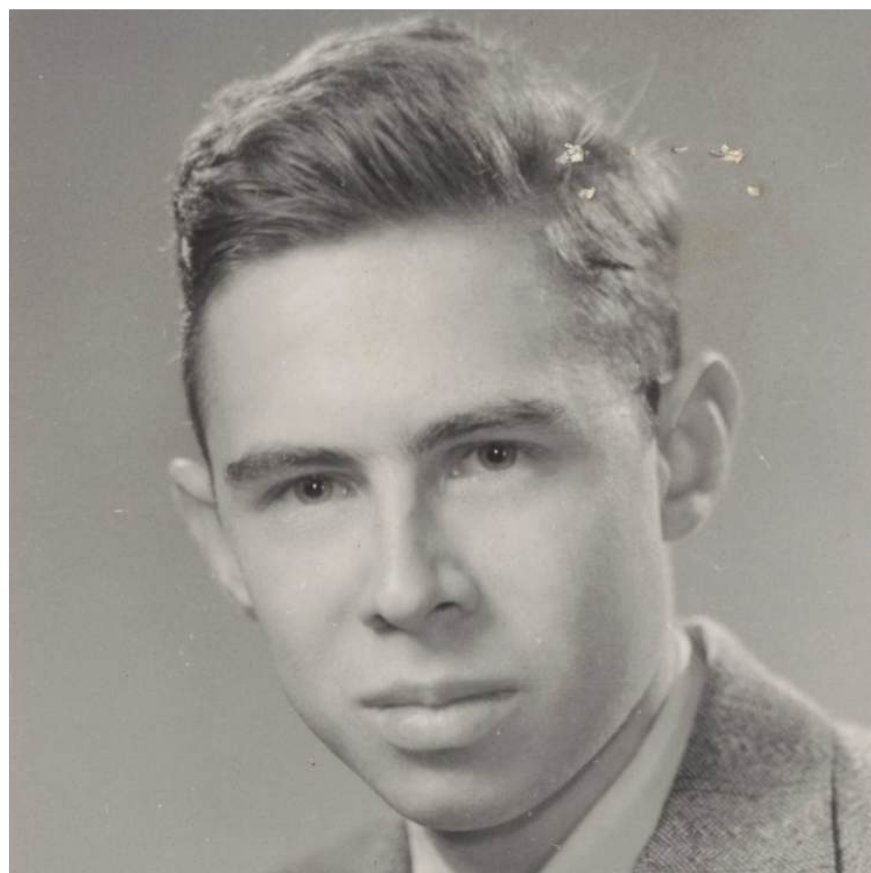
Arthur Samuel

Checkers Program, 1952

Created a Checkers-playing program that got better overtime.



Also introduced the term
“Machine Learning.”

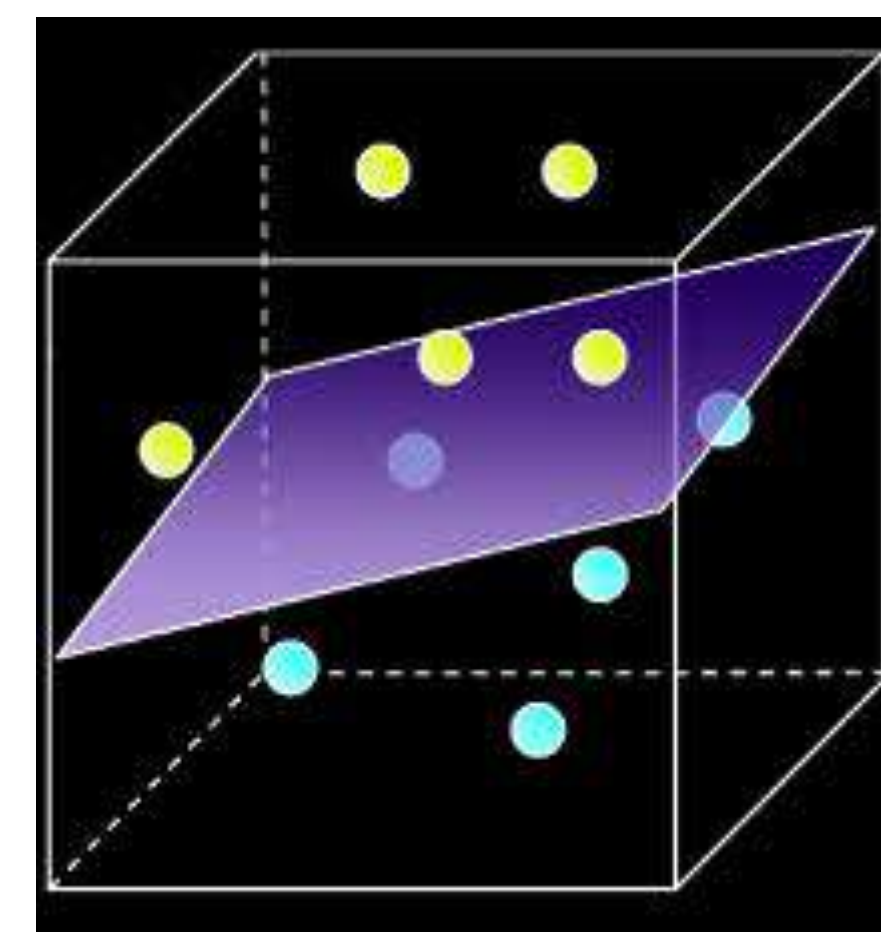
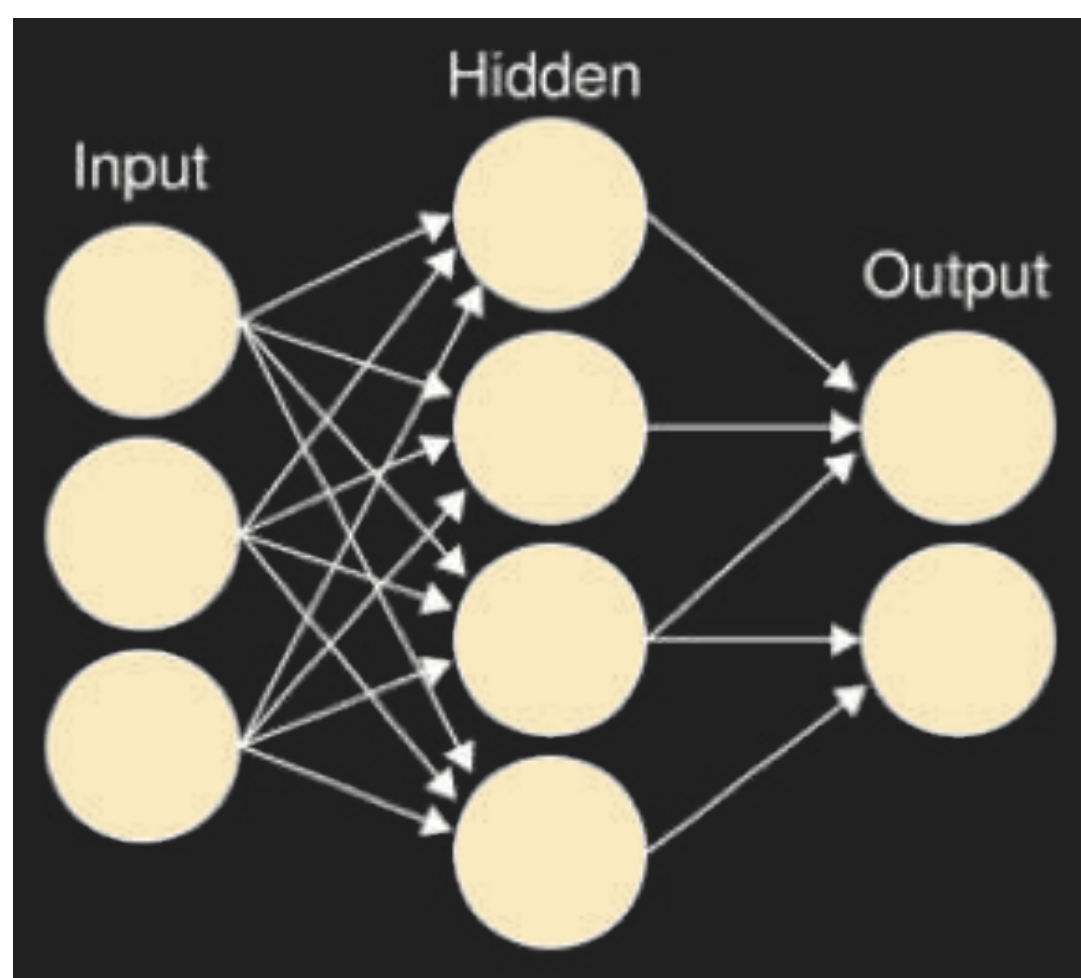


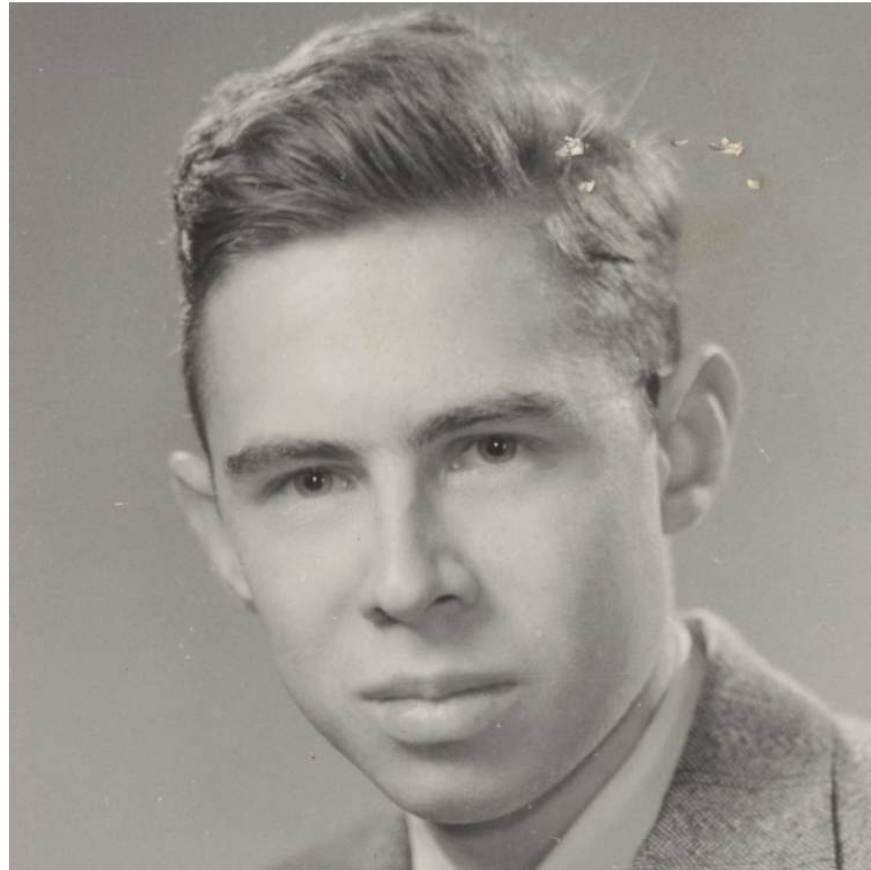
Frank Rosenblatt
@ Cornell!

Perceptron, 1957

Predecessor of deep networks.

Separating two classes of objects using a linear threshold classifier.





Frank Rosenblatt
@ Cornell!

NEW NAVY DEVICE LEARNS BY DOING

Psychologist Shows Embryo
of Computer Designed to
Read and Grow Wiser

WASHINGTON, July 7 (UPI)—The Navy revealed the embryo of an electronic computer today that it expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.

The embryo—the Weather Bureau's \$2,000,000 "704" computer—learned to differentiate between right and left after fifty attempts in the Navy's demonstration for newsmen.

The service said it would use this principle to build the first of its Perceptron thinking machines that will be able to read and write. It is expected to be finished in about a year at a cost of \$100,000.

Dr. Frank Rosenblatt, designer of the Perceptron, conducted the demonstration. He said the machine would be the first device to think as the human brain. As do human beings, Perceptrons will make mistakes at first, but will grow wiser as it gains experience, he said.

Dr. Rosenblatt, a research psychologist at the Cornell Aeronautical Laboratory, Buffalo, said Perceptrons might be fired to the planets as mechanical space explorers.

Without Human Controls

The Navy said the perceptron would be the first non-living mechanism "capable of receiving, recognizing and identifying its surroundings without any human training or control."

The "brain" is designed to remember images and information it has perceived itself. Ordinary computers remember only what is fed into them on punch cards or magnetic tape.

Later Perceptrons will be able to recognize people and call out their names and instantly translate speech in one language to speech or writing in another language, it was predicted.

Mr. Rosenblatt said in principle it would be possible to build brains that could reproduce themselves on an assembly line and which would be conscious of their existence.

In today's demonstration, the "704" was fed two cards, one with squares marked on the left side and the other with squares on the right side.

Learns by Doing

In the first fifty trials, the machine made no distinction between them. It then started registering a "Q" for the left squares and "O" for the right squares.

Dr. Rosenblatt said he could explain why the machine learned only in highly technical terms. But he said the computer had undergone a "self-induced change in the wiring diagram."

The first Perceptron will have about 1,000 electronic "association cells" receiving electrical impulses from an eye-like scanning device with 400 photo-cells. The human brain has 10,000,000,000 responsive cells, including 100,000,000 connections with the eyes.

The New York Times

Published: July 8, 1958

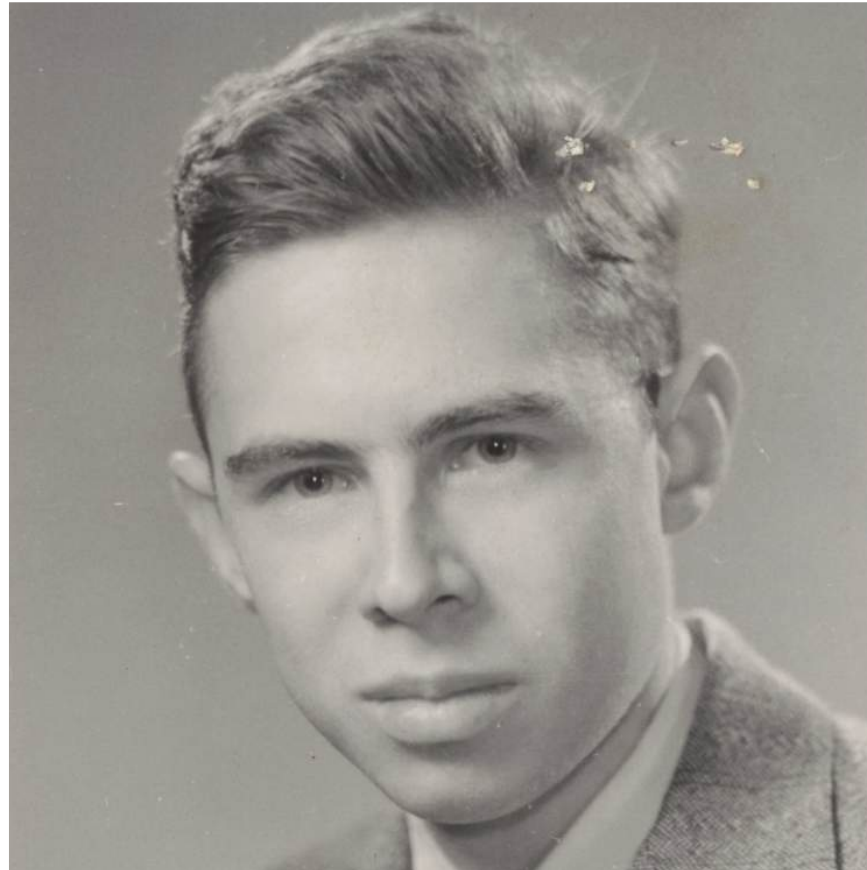
Copyright © The New York Times

Perceptron, 1957

New Navy Device Learns by Doing

- The New York Times (July 8, 1958)

“Later perceptrons will be able to recognize people and call out their names and instantly translate speech in one language to speech or writing in another language, it was predicted.”



Frank Rosenblatt
@ Cornell!

Neural Network Boom, 1960-79

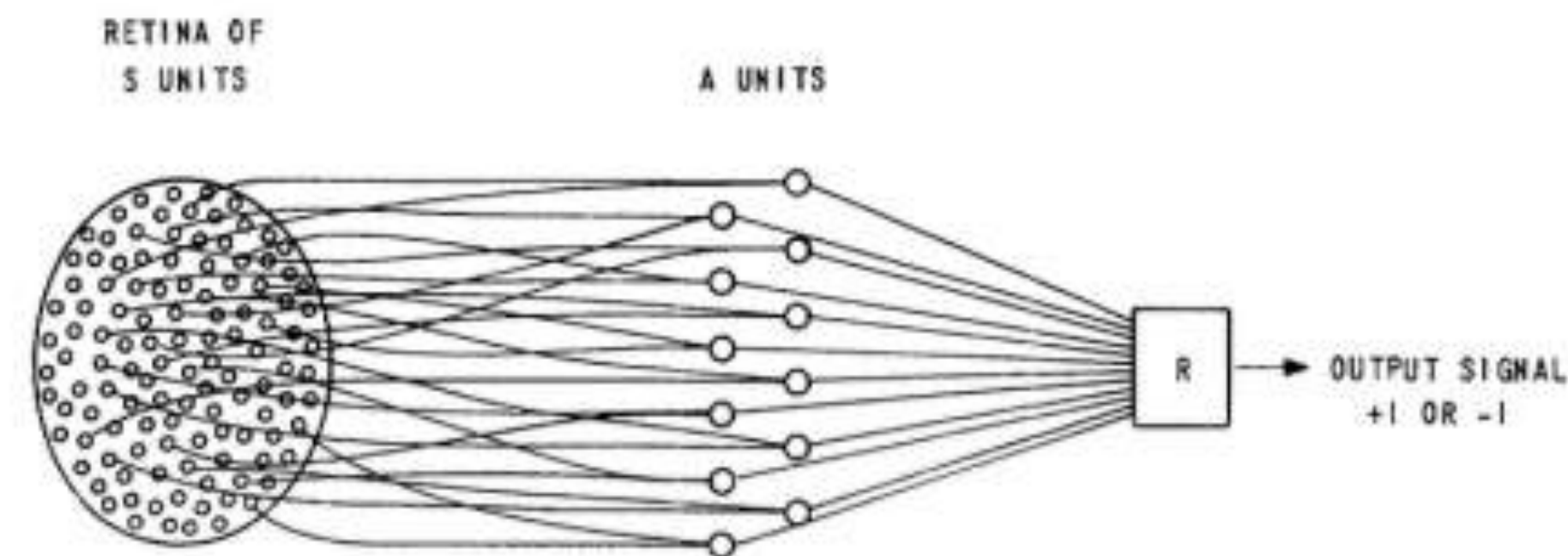


Fig. 2. Network organization of a typical elementary perceptron

SEPTEMBER 1964

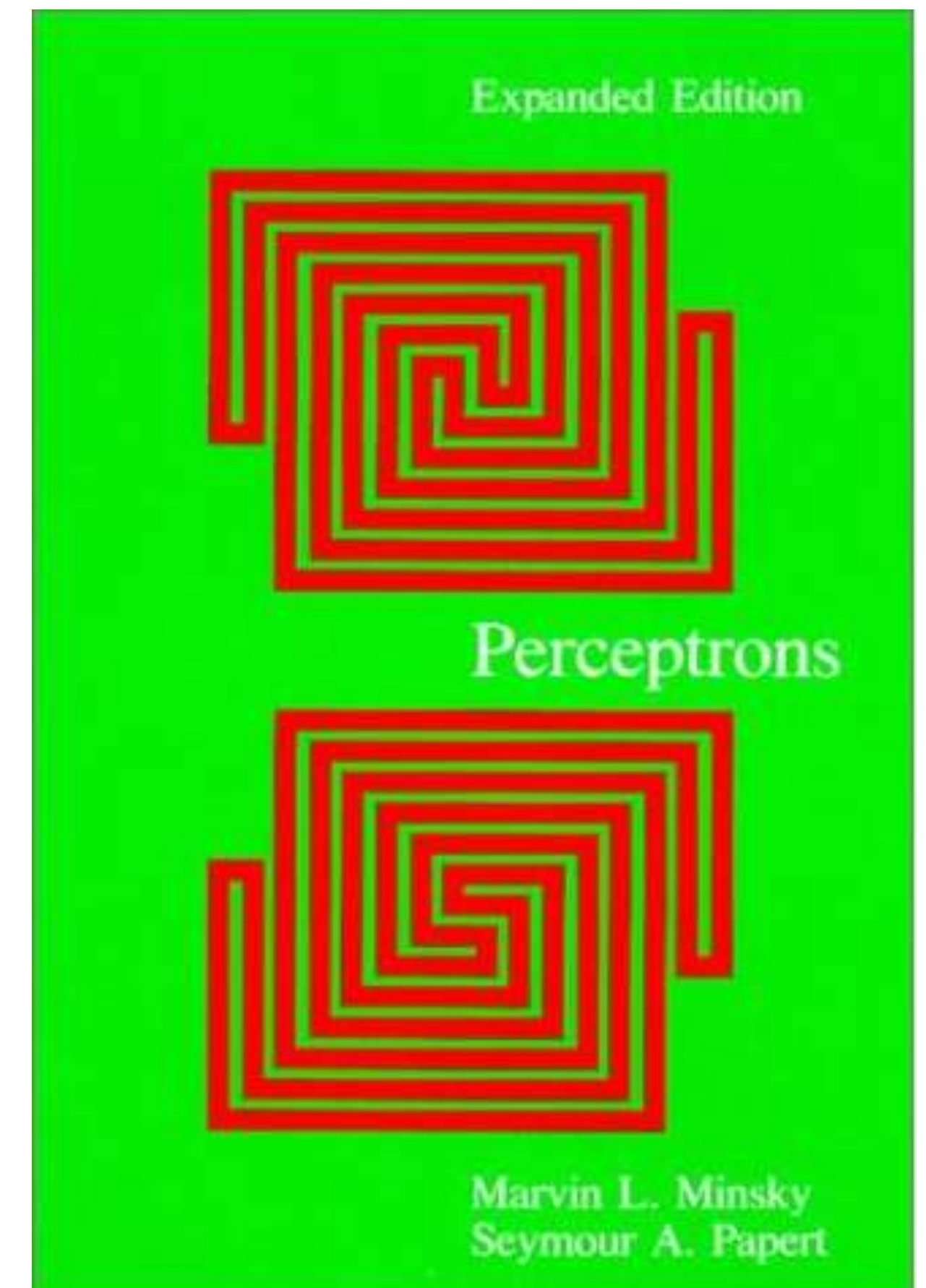
Rosenblatt—Analytic Techniques for the Study of Ne

- 1962 Rosenblatt invents Multi-Layer Perceptron (MLP) (fixed hidden layer)
- 1965 Ivakhnenko and Lapa introduce first Feed Forward Neural Net (FFNN)
- 1967 First FFN trained with SGD [Amari 1967]
- 1970 Modern back-propagation is introduced by Seppo Linnainmaa

AI Winter (1974-1980)

- (1969) Minsky & Papert “killed” AI
 - simple XOR dataset
- Burst huge expectation bubble
- Speech understanding / translation fails
- 1973 Lighthill report:

"In no part of the field have the discoveries made so far produced the major impact that was then promised"
- UK and US stop funding AI research

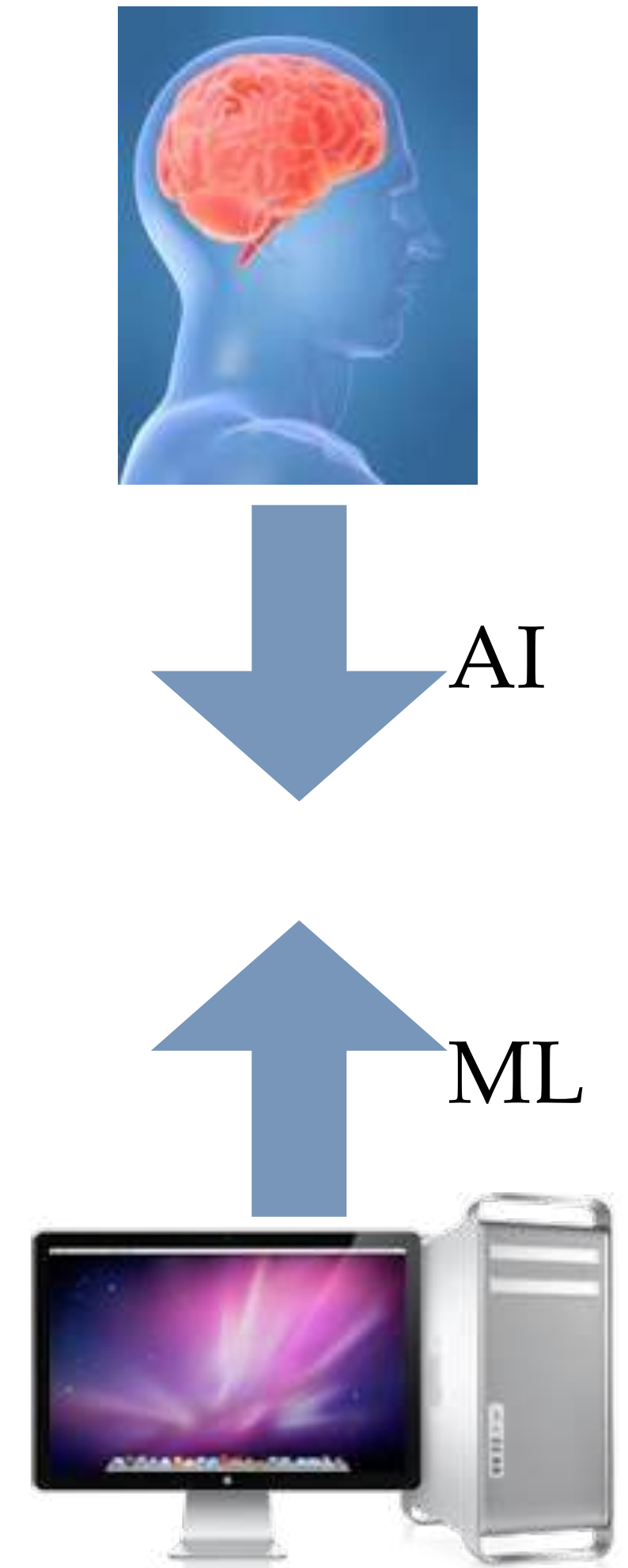


AI Boom then ML

- AI Boom: 1980-87
 - Search Algorithms (Eg. A*)
 - Hard-coded, rule-based "Expert Systems"
 - Symbolic reasoning, logic, Lisp machines
- AI Winter (1987-93)
 - Expectations overhyped (again)

Rebirth as Machine Learning

- Machine Learning:
 - Originally: Mostly a name game to get funding.
- Profound difference:
 - ML: Bottom up, AI: Top down
 - ML: More practical smaller goals
 - Based on **Statistics and Optimization, not Logic**



TD-Gammon (1994)

- Gerry Tesauro (IBM) teaches a neural network to play Backgammon.
- The net plays 100K+ games **against itself** and beats world champion [Neurocomputation 1994]
- Algorithm found new techniques that people had erroneously ruled out.



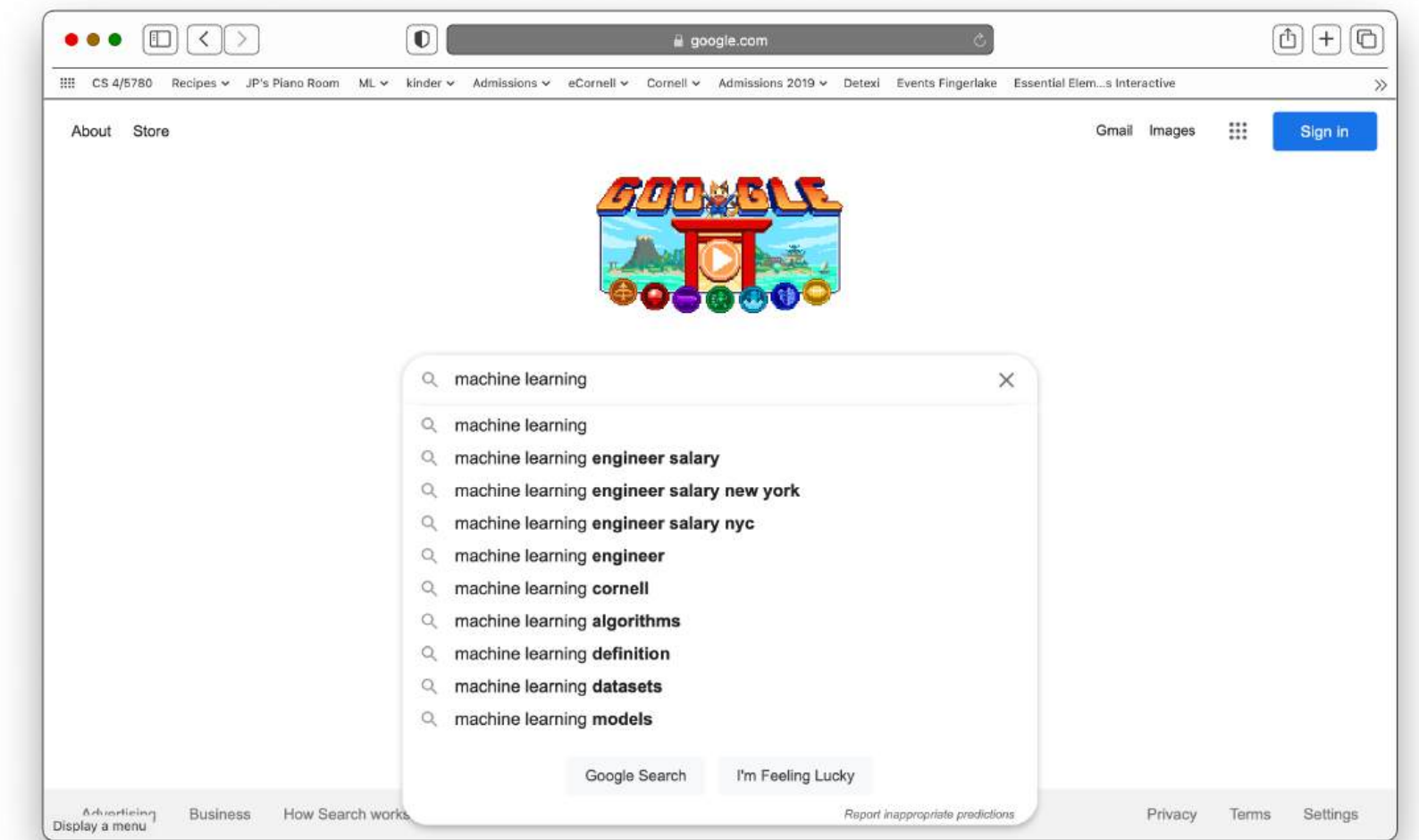
Deep Blue (1997)

- IBM's Deep Blue wins against Kasparov in chess. Crucial winning move is made due to Machine Learning (G. Tesauro).
 - (Mostly a more classical AI system)



Expanding the reach, 2000s

- Learning to rank
 - Powering search engines: Google, Bing, ...
- Topic Modeling
 - Detecting and organizing documents by subject matter.
 - Making sense of the unstructured data on the web.
- Online economy:
 - Ad placement and pricing.
 - Product recommendation.



\$\$\$

Machine learning became profitable!

2000-2010 The Convex Years

- Success of Support Vector Machines (SVM) and Convex Optimization
- Kernelization enabled non-linear classifiers that still use convexity
- Rigorous mathematical guarantees with empirical success
- Led to development of optimization algorithms that are workhorse of current models

Datasets and Empiricism, 2010s

- Large-scale datasets like ImageNet: 14,197,122 images with 21,841 distinct class labels.
- Community competitions to improve upon methods: e.g., the ImageNet Large Scale Visual Recognition Challenge (ILSVRC).



Fei-Fei Li



Return of Neural Networks, 2010s

Neural networks return and excel at image recognition, speech recognition, ...

The 2018 Turing award was given to Yoshua Bengio, Geoff Hinton, and Yann LeCun.



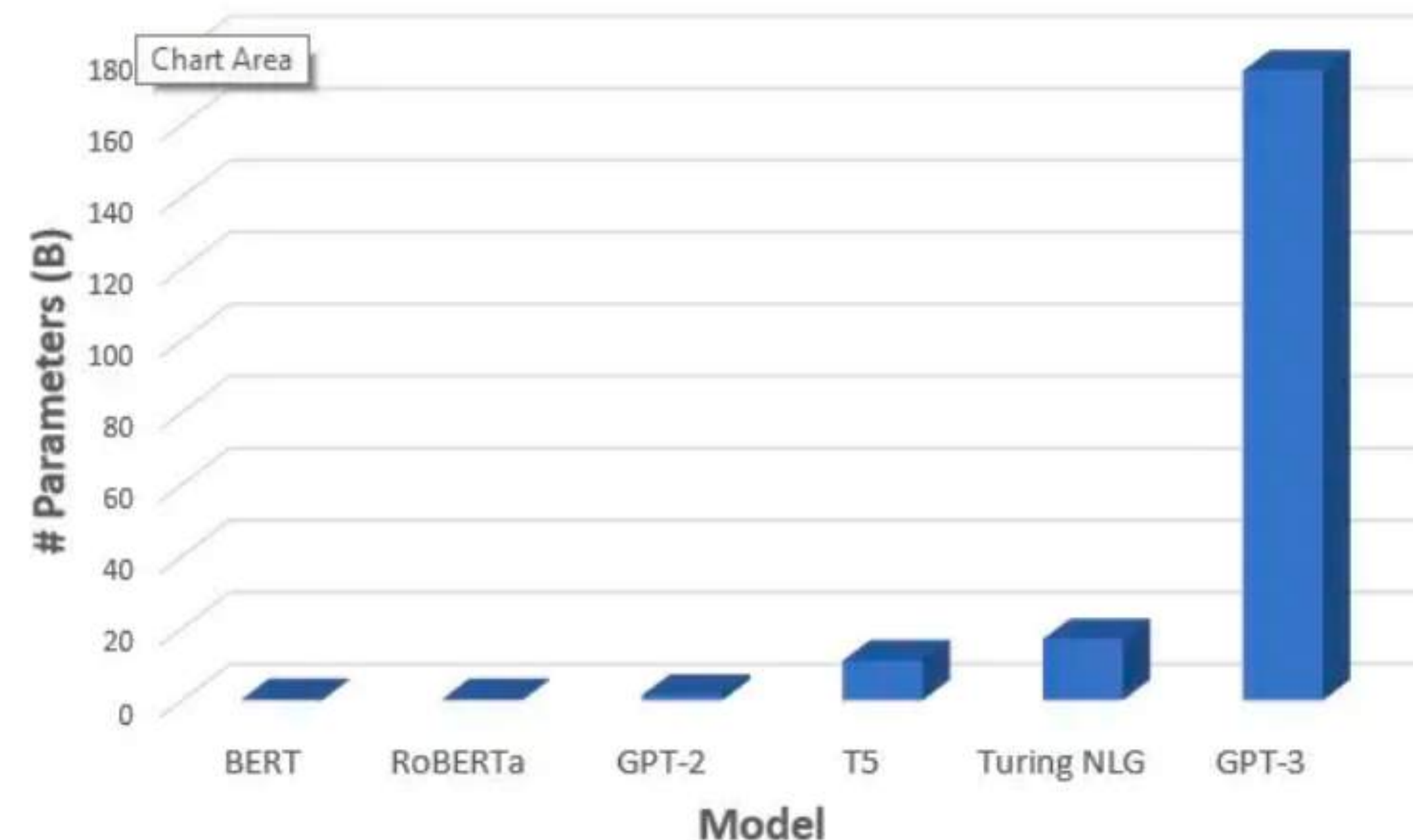
2016 Alpha Go

- 1920 CPUs and 280 GPUs
- Deep Mind's Alpha Go wins against Lee Sedol 5:1
- Big shock
 - China invests heavily in AI research
- Beginning of “AI arms race”



2021 Generative Models become Mainstream

- 178 Billion Parameters
- \$4.6 Million estimated cost
- 355 GPU Years of Training
- Trained on 45TB of text



ML is Creative (well maybe we are still working on it)

ChatGPT ▾

write a funny rap song on



Sure! What topic do you want the rap song to be about?

diffusion models



"Diffusion Beats (The AI Flow)"

[Intro]

Yo, it's your boy AI, spittin' machine-made bars,
Talkin' 'bout diffusion models takin' over like stars.
From noise to clarity, it's how we roll,
Let me break it down, control to goal.

[Verse 1]

Started with some noise, yeah, a cloud of confusion,
Pixel party wild, that's the initial diffusion.
Addin' noise forward, things get outta sync,
But then we reverse it—watch the data rethink.

Models spinnin' like a DJ, crafting the beat,
From static to masterpiece, can't be beat.

GANs tried their shot, but we took the crown

Diffusion's the king, wearin' that AI gown.



And Versatile

ChatGPT ▾

make a funny picture that describes machine learning as programs that improve with experience



& 2024

Two 2024 Nobel Prizes Honor Researchers Innovating in AI, Machine Learning

October 21, 2024



Recognizing the transformative power of AI, the Royal Swedish Academy of Sciences recently awarded the 2024 Nobel Prizes in physics and chemistry to researchers at the forefront of machine learning, artificial neural networks, computational design, and protein structure prediction.

John Hopfield and Geoffrey Hinton earned the [Nobel Prize in physics](#) for “foundational discoveries and inventions that enable machine learning with artificial neural networks.” David Baker, Demis Hassabis, and John Jumper received the [Nobel Prize in chemistry](#) for advances in computational protein design and protein structure prediction.

ML For Astronomy

ScienceNews
INDEPENDENT JOURNALISM SINCE 1921

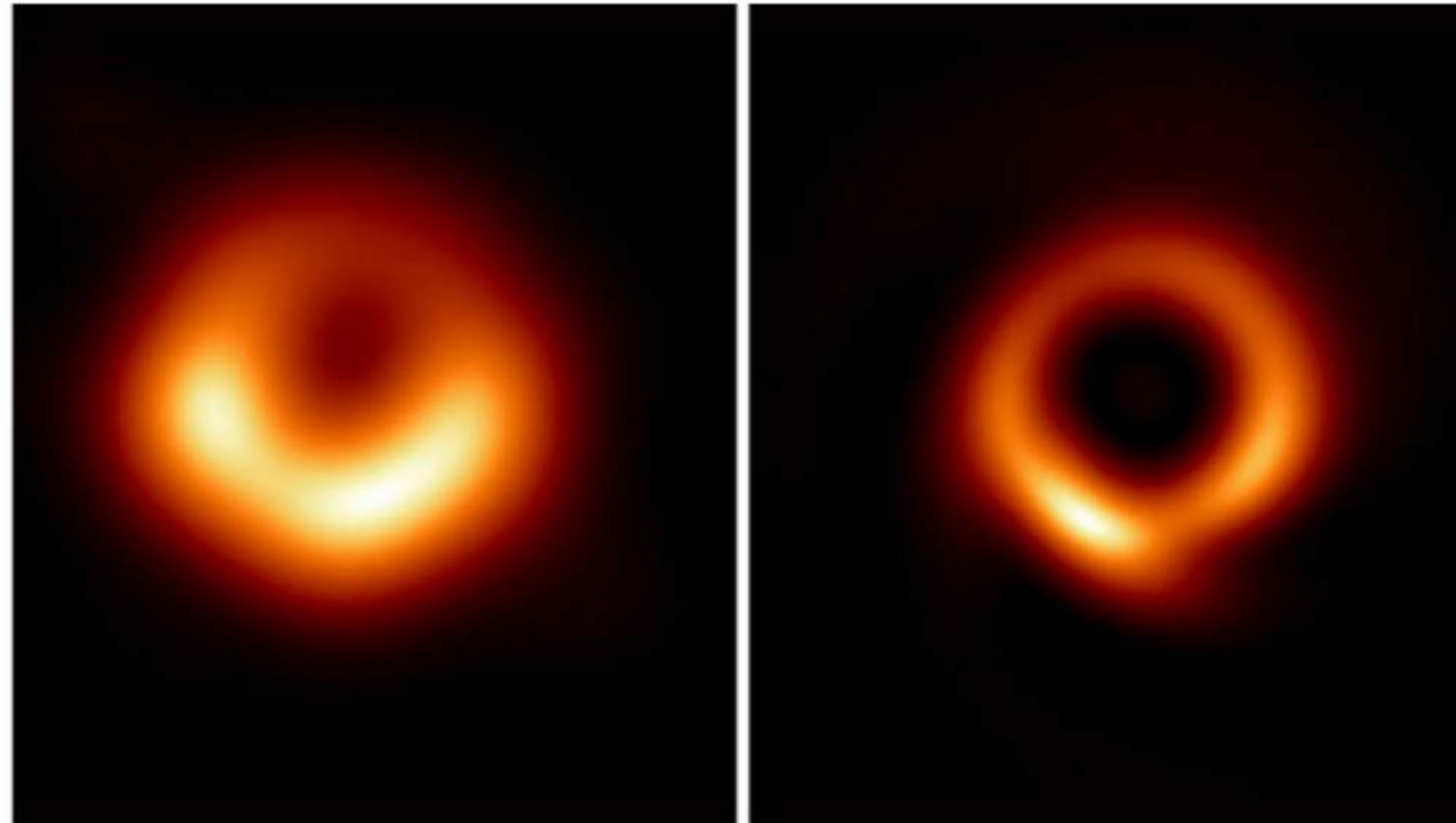
ALL TOPICS HEALTH HUMANS LIFE EARTH PHYSICS SPACE

NEWS

ASTRONOMY

The first black hole portrait got sharper thanks to machine learning

The technique could help scientists get a better handle on the behemoth's mass



Scientists used machine learning to create a sharper image of the supermassive black hole in the galaxy M87 (right), compared with the original image (left).

L. MEDEIROS ET AL 2023

ML in Material Sciences

THE CONVERSATION

Academic rigor, journalistic flair

Search analysis, research, academics...



Arts + Culture Economy Education Environment + Energy Ethics + Religion Health Politics + Society Science + Tech World Podcasts Local



AI is transforming the search for new materials that can help create the technologies of the future

Published: February 10, 2025 7:56am EST

Battery technology is one area that can benefit from the development of novel materials. IM Imagery / Shutterstock

AI for Mathematics

MATHEMATICS + AI

Recent research-level mathematical breakthroughs

Search research...

News

Research

Open Problems

Formal Proof

About

AI is beginning to crack long-standing open problems in mathematics

What is actually solved — and what is still only progress

2026	SOLVED	Erdős unit-distance conjecture	— OpenAI model disproved the nearly 80-year-old conjecture.
2026	SOLVED	Jacobian conjecture (dimension ≥ 3)	— Claude Fable 5 helped find an explicit counterexample in $\mathbb{C}^3$; the 2D case remains open.
2026	SOLVED	9 open Erdős problems	— Google DeepMind formal-proof agent resolved 9 of 353 problems with Lean-verified proofs.
2025	PROGRESS	Andrews–Curtis conjecture	— AI ruled out two ~25-year-old candidate-counterexample families; the main conjecture is open.
2023	PROGRESS	Cap set problem	— FunSearch discovered new record constructions; the full extremal problem remains open.
2026	FORMALIZED	Random Bernoulli matrices	— GPT-5.6 Sol + Claude Opus 5 formalized Tikhomirov’s 2020 theorem in Lean — not a new solution.

Sources: OpenAI (May 2026) • Anthropic / MathOverflow (Jul 2026) • DeepMind / arXiv:2605.22763 • Caltech (2025) • DeepMind FunSearch (2023) • LeanEval (Aug 2026)

Outlook



“A breakthrough in machine learning would be worth ten Microsofts.” (Bill Gates, Microsoft)

“It will be the basis and fundamentals of every successful huge IPO win in 5 years.” (Eric Schmidt, Google / Alphabet)



“AI and machine learning are going to change the world and we really have not begun to scratch the surface.”
(Jennifer Chayes, UC Berkeley)

“ML is transforming sector after sector of the economy, and the rate of progress only seems to be accelerating.” (Daphne Koller, Stanford / Coursera/ Insitro)



“Machine learning is the next Internet” (Tony Tether, DARPA)

DANGER



Data privacy / misuse

Learning models leak training data
(Fredrickson et al. '15)



Leaked data

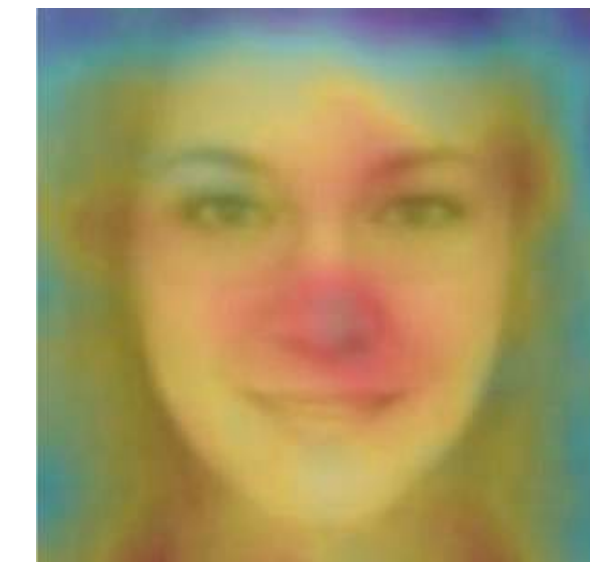


Real image

Formal definitions of data privacy:

- **K- anonymity (Sweeney)**
- **Differential Privacy (Dwork)**

Learning algorithms detect sexual orientation
better than people
(Wang & Kosinski'17)



Latanya Sweeney



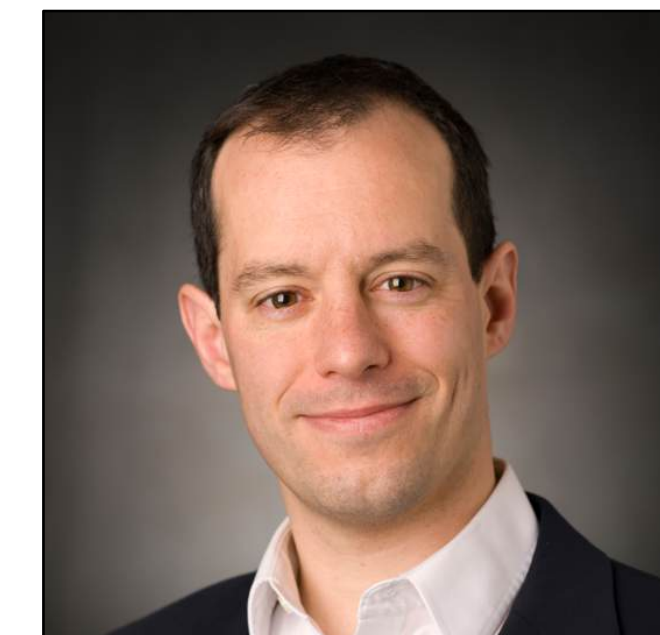
Cynthia Dwork



Frank McSherry



Kobbi Nissim



Adam Smith

Robust and Secure ML

Image Recognition

Misreading traffic signs

(Eykholt et al)



Speech recognition

Hide commands in noise (Carlini & Wagner)



Poisoning Attacks

Tay (chat bot) became inflammatory in 16 hr.



How to create robust and secure machine learning algorithms?

Learning and the Society

- Bad dynamics perpetuate and worsen stereotypes and biases.
- Who carries the burden of bad prediction?
- How to design good dynamics?

The Best Algorithms Struggle to Recognize Black Faces Equally

Google's algorithm shows prestigious job ads to men, but not to women. Here's why that should worry you.

Gender and racial bias found in Amazon's facial recognition technology (again)

How Amazon Accidentally Invented a Sexist Hiring Algorithm

A company experiment to use artificial intelligence in hiring inadvertently favored male candidates.

Do Google's 'unprofessional hair' results show it is racist?



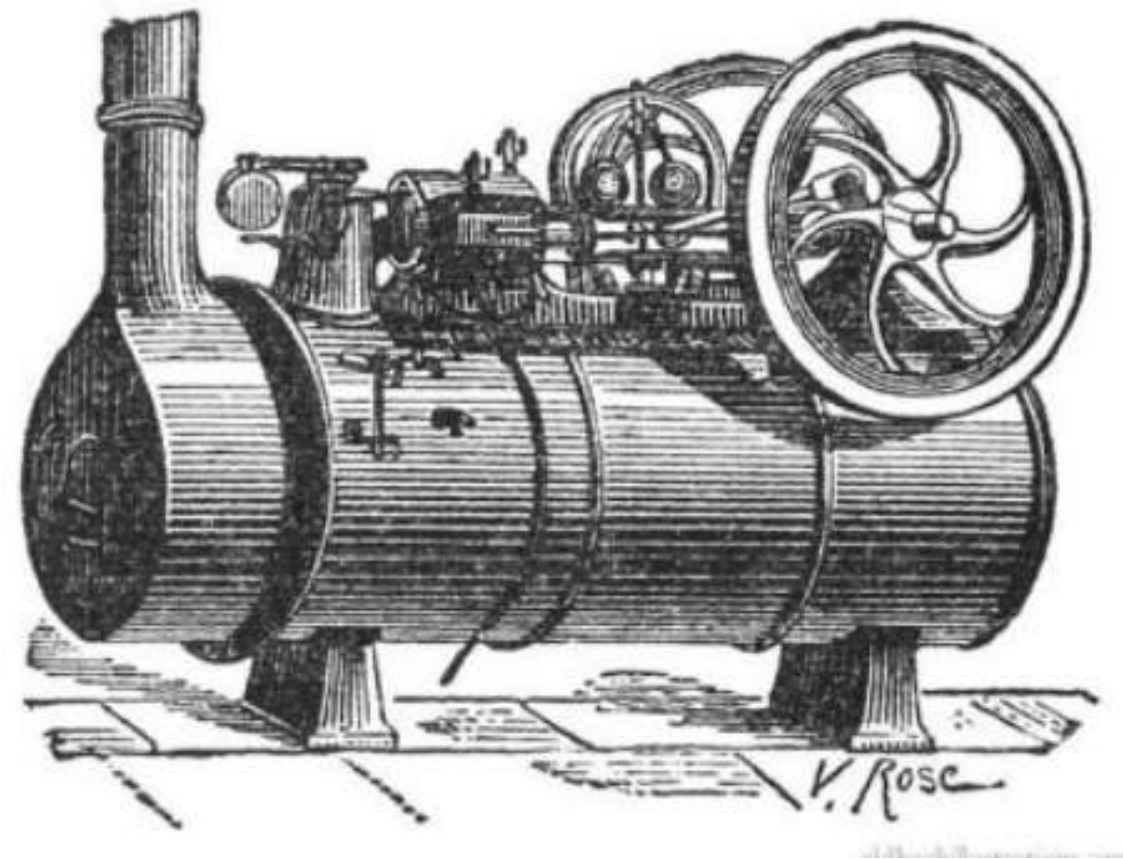
When an Algorithm Helps Send You to Prison

By Ellora Thadaney Israni



ML in Society

- New technology that can transform society can also overwhelm it
 - Steam engine late 1800s (lead to sweat shops, exploitation)
 - Cars: out-of-control traffic accidents until 1970s



To-do action items

- Pass Placement Exam!
- Find study group
- Autograder Setup:
 - Find Teammate
 - Start Project -1
 - Start Project 0